AS102-A009

River Discharge and Ocean Spin-Up Effects on Coupled Regional Climate Simulations Over CORDEX East Asia

Junseo PARK+, Woojin CHO, Seok-Woo SHIN, Dong-Hyun CHA#

Ulsan National Institute of Science and Technology (UNIST)

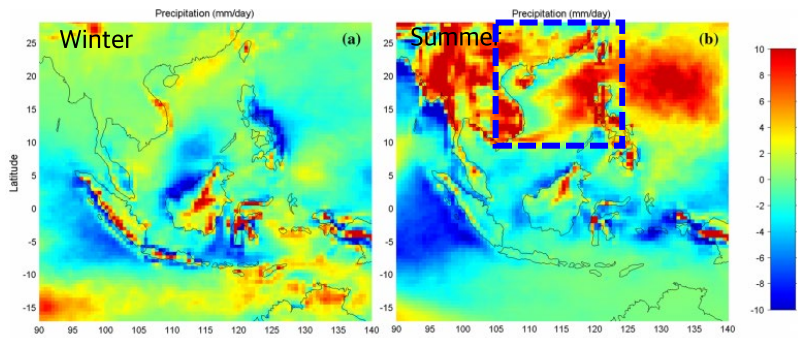
Contents

1. **Introduction**
2. **Data & Methods**
3. **Results**
    1. Coupled Effect
    2. Ocean Spin-up Effect
    3. River Discharge Effect
4. **Summary**

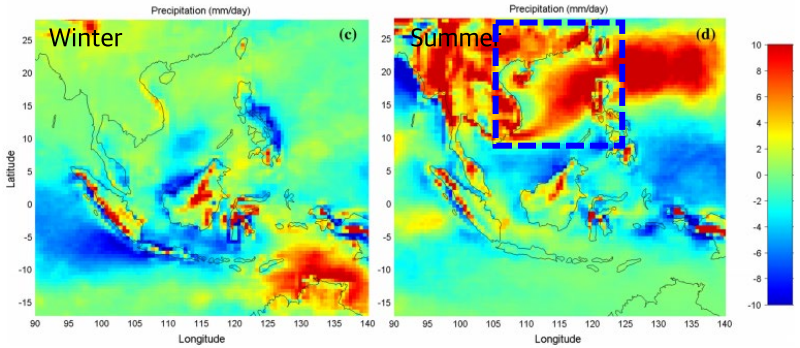
1. Introduction

1-1. Motivation: Why Coupling?

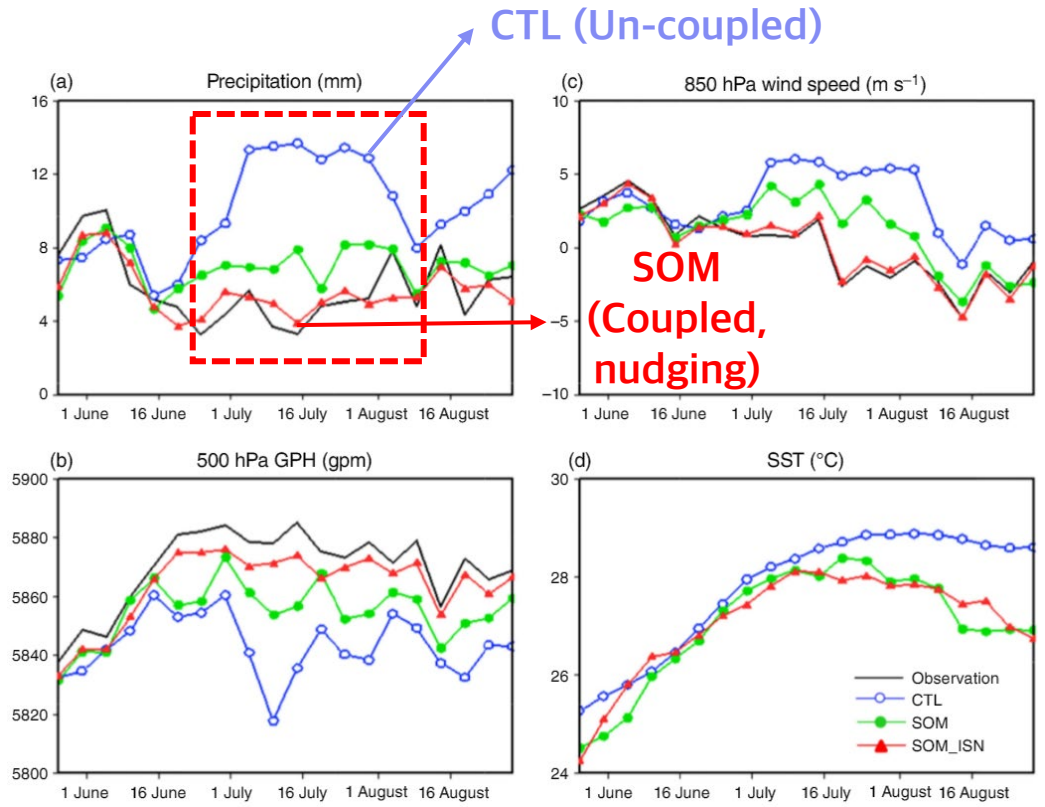
Coupled RCM (RegCM3 & SOM)



RegCM3-only



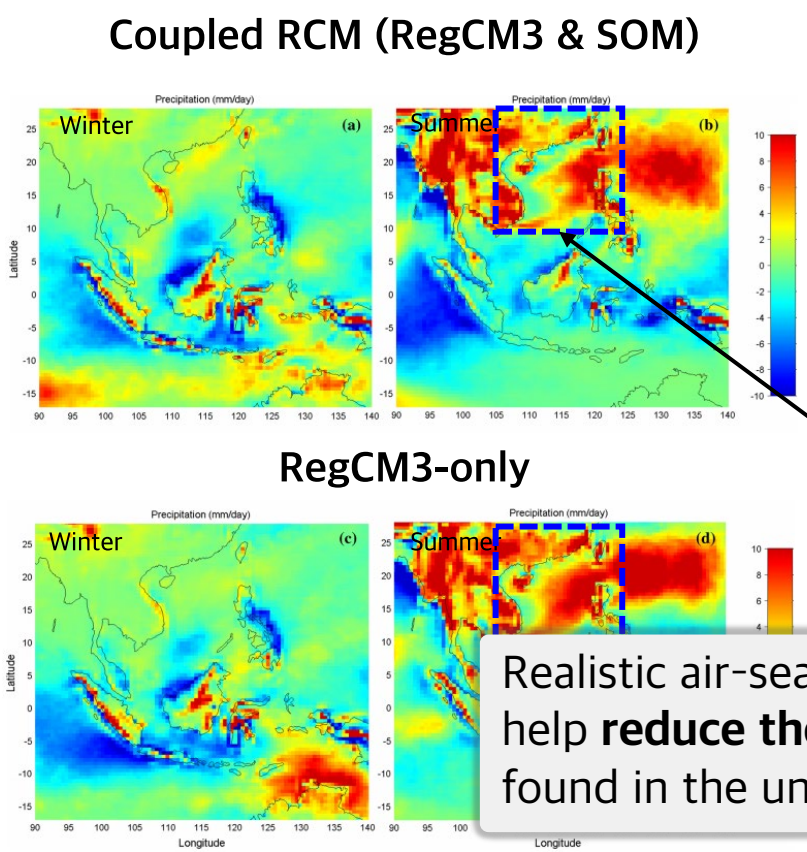
Bias of Coupled Regional Climate Model (RCM) and uncoupled RCM
(Wei et al., 2013)



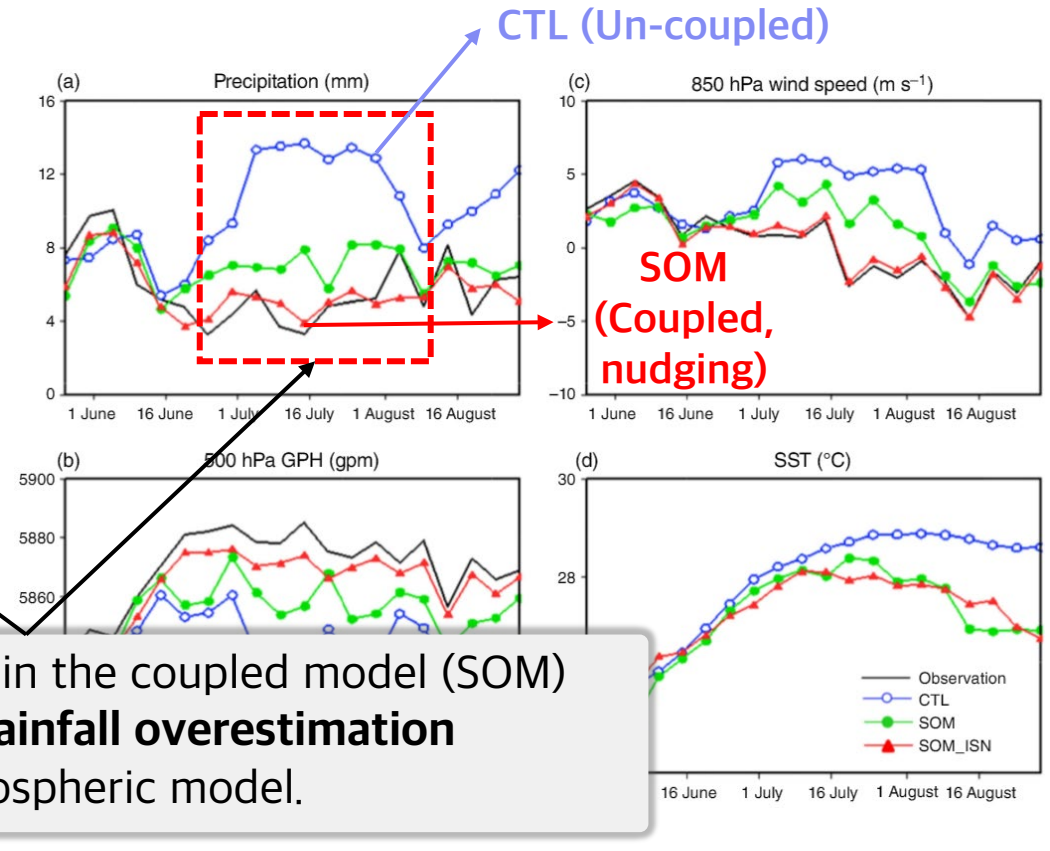
Time series of simulations from the Coupled and the uncoupled RCM
(Cha et al., 2015)

RegCM3: Regional Climate Model version 3
SOM: Slab Ocean Model
RCM: Regional Climate Model

1-1. Motivation: Why Coupling?



Bias of Coupled Regional Climate Model (RCM) and uncoupled RCM
(Wei et al., 2013)



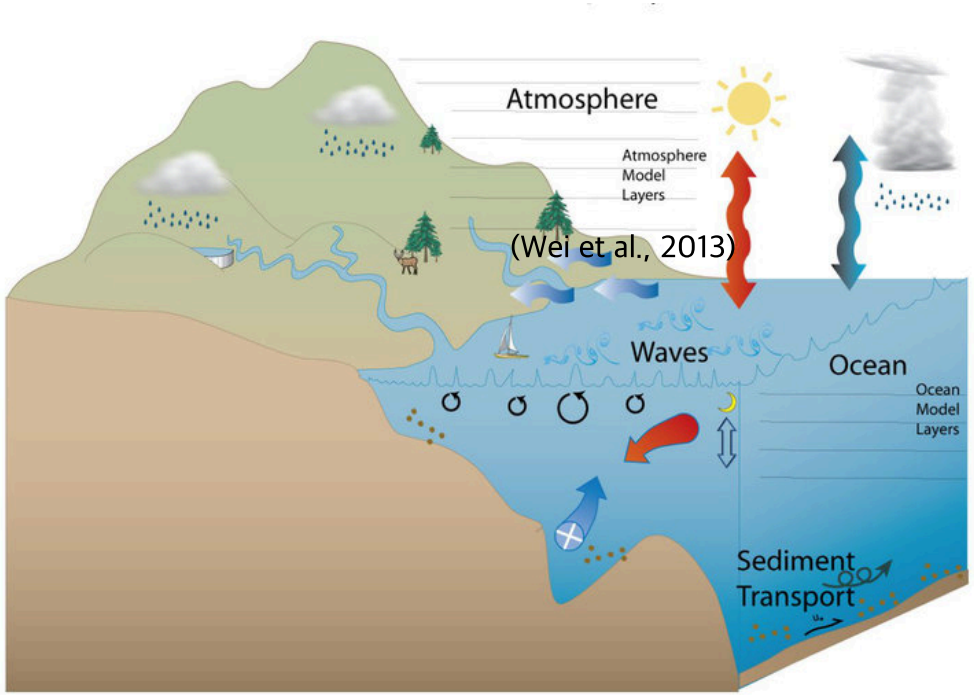
Time series of simulations from the Coupled and the uncoupled RCM
(Cha et al., 2015)

Realistic air-sea interactions in the coupled model (SOM) help **reduce the monsoon rainfall overestimation** found in the uncoupled atmospheric model.

RegCM3: Regional Climate Model version 3
SOM: Slab Ocean Model
RCM: Regional Climate Model

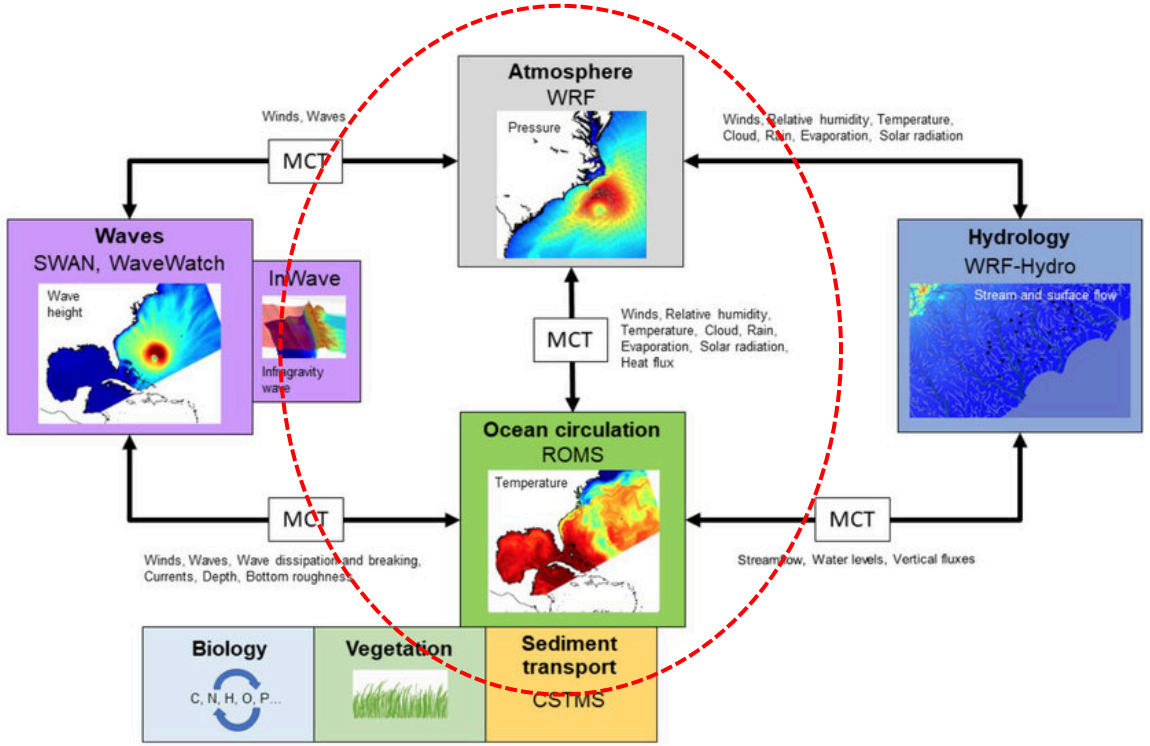
1-2. Numerical Model

Coupled Ocean Atmosphere Wave Sediment Transport (COAWST)



(Wei et al., 2013)

COAWST Modeling System infographic
By Coastal and Marine Hazards and Resources Program (2015)



COAWST Modeling System
By Woods Hole Coastal and Marine Science Center

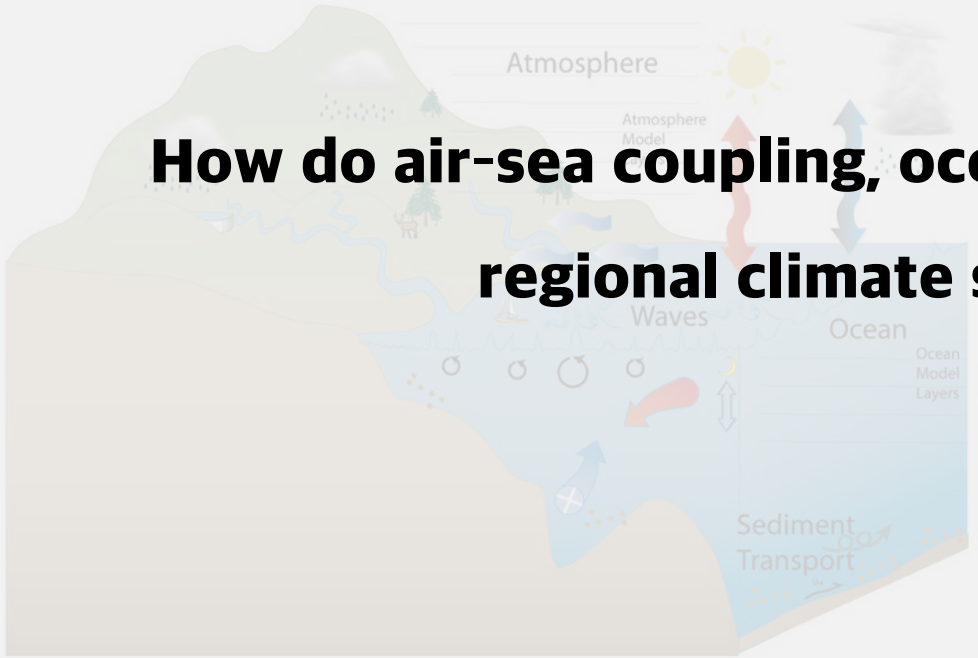
(Warner et al., 2010)

WRF: Weather Research and Forecast
ROMS: Regional Ocean Modeling System
MCT: Model Coupling Toolkit

1-2. Numerical Model

Coupled Ocean Atmosphere Wave Sediment Transport (COAWST)

How do air-sea coupling, ocean spin-up, and river discharge affect regional climate simulations over East Asia?



COAWST Modeling System infographic
By Coastal and Marine Hazards and Resources Program (2015)

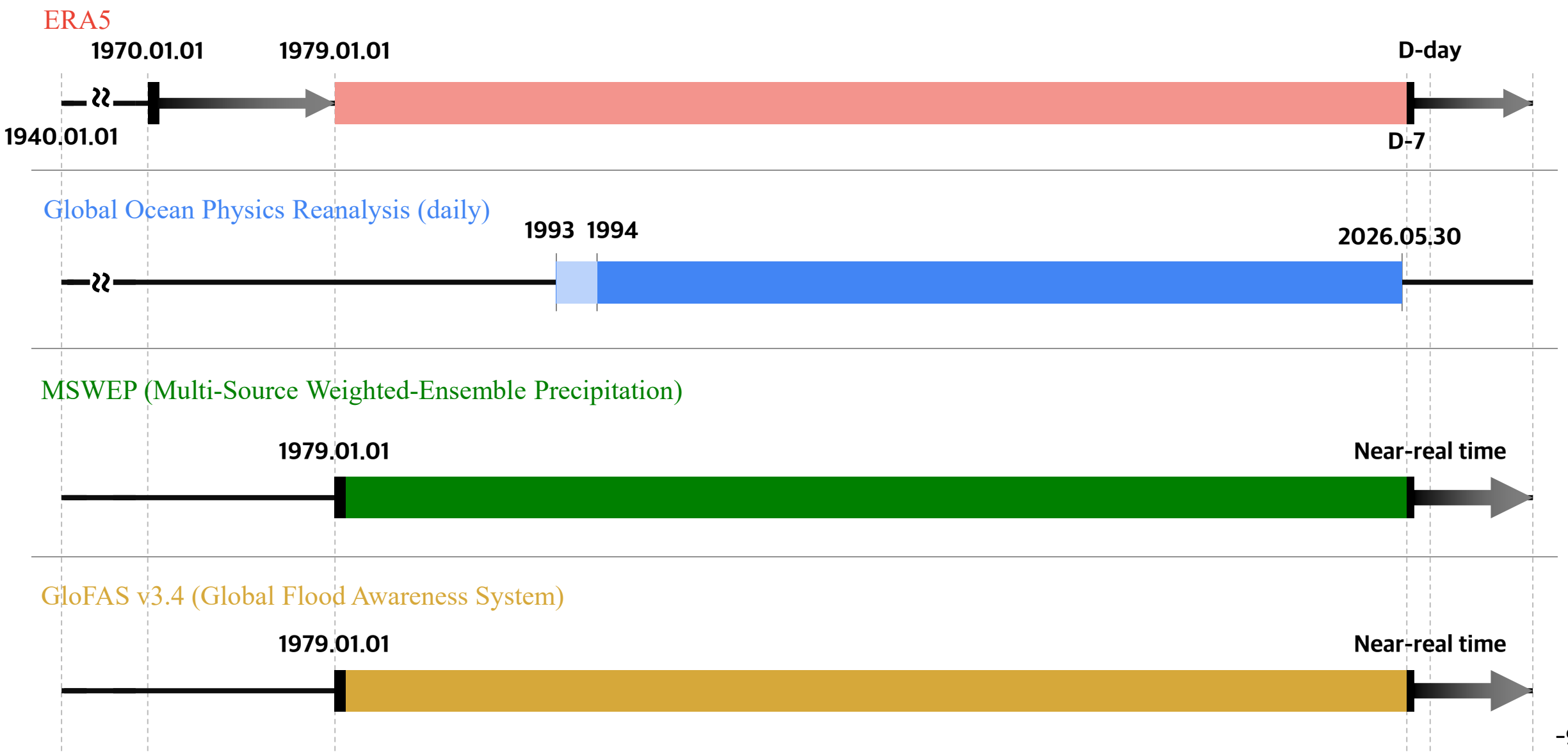


COAWST Modeling System
By Woods Hole Coastal and Marine Science Center

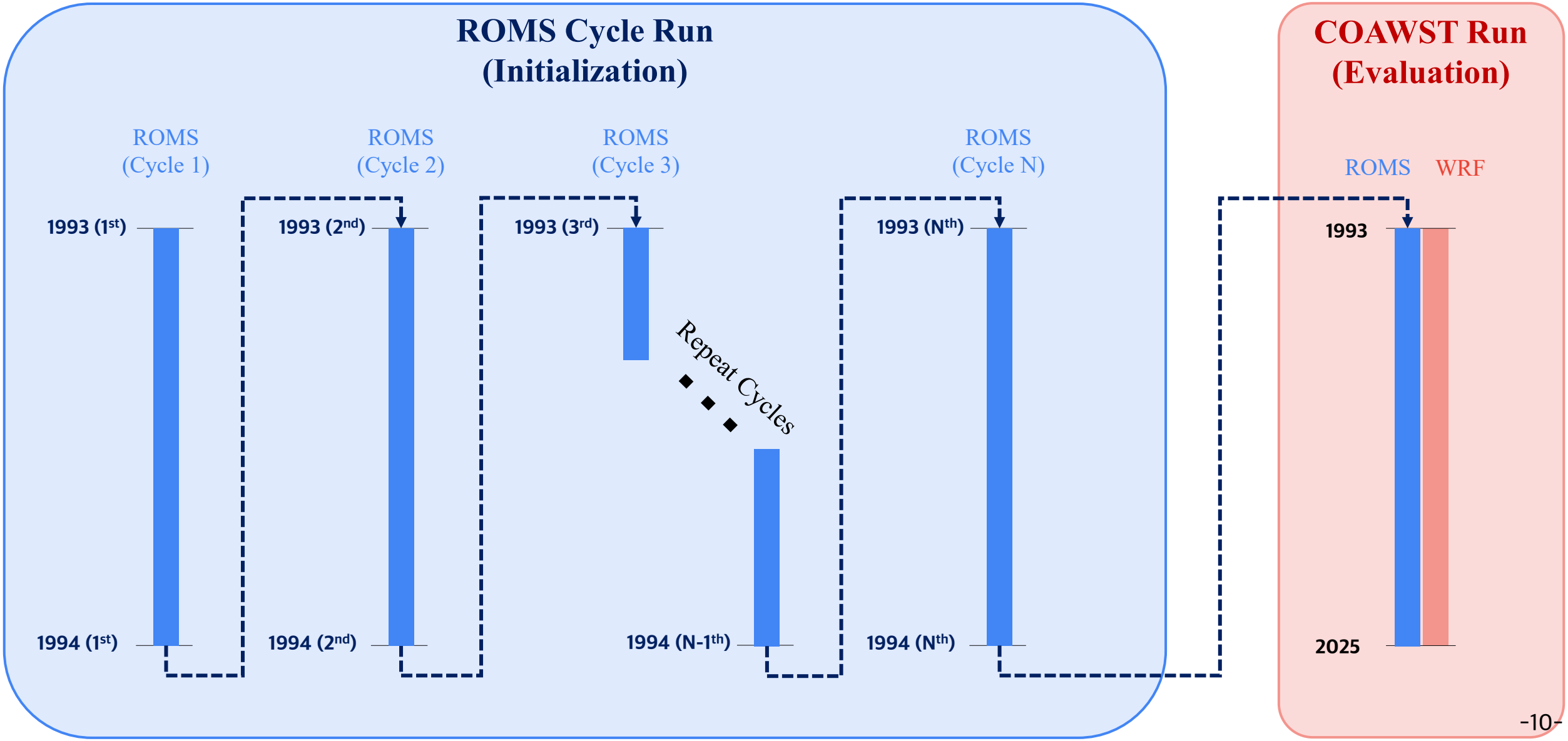
WRF: Weather Research and Forecast
ROMS: Regional Ocean Modeling System
MCT: Model Coupling Toolkit

2. Data & Method

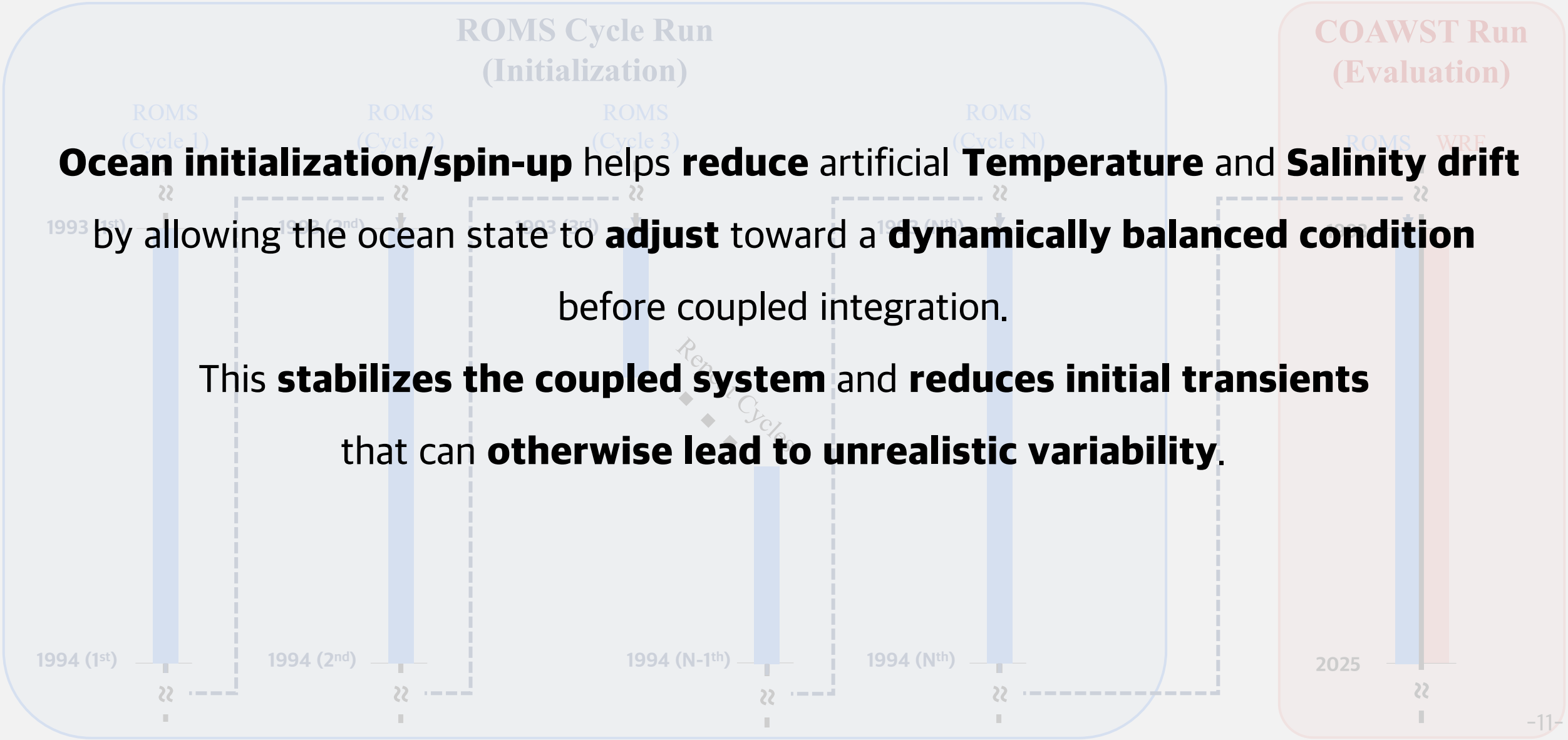
2-1. Data


ERA5
1970.01.01
1979.01.01
D-day
1940.01.01
D-7
Global Ocean Physics Reanalysis (daily)
1993
1994
2026.05.30
MSWEP (Multi-Source Weighted-Ensemble Precipitation)
1979.01.01
Near-real time
GloFAS v3.4 (Global Flood Awareness System)
1979.01.01
Near-real time

2-2. Ocean Initialization Method



2-2. Ocean Initialization Method



Ocean initialization/spin-up helps **reduce** artificial **Temperature** and **Salinity drift** by allowing the ocean state to **adjust** toward a **dynamically balanced condition** before coupled integration.

This **stabilizes the coupled system** and **reduces initial transients** that can **otherwise lead to unrealistic variability**.

2-3. Model Configuration

Couple System	Coupled Ocean-Atmosphere-Wave-Sediment Transport (COAWST)
Coupler	Model Coupling Toolkit (MCT) v 2.6.0
Ocean	Regional Ocean Modeling System (ROMS) 3.9
Atmosphere	Weather Research and Forecasting Model (WRF) v 4.2.2
Wave(s)	Simulating Waves Nearshore (SWAN) v 41.31 WAVEWATCH III (WW3) v 7.12 Infragravity wave model (InWave) v 2.0
Sediment Transport	Community Sediment Transport Modeling Systems (CSTMS)

Atmosphere	Weather Research and Forecasting Model (WRF) v 4.2.2
Horizontal Resolution | (Domain)	12 km (825×521) / dt = 36 s
Vertical Layers	35 hybrid levels
Microphysics	WSM6
Cumulus Parameterization	MSKF
Radiation	RRTMG
Planetary Boundary Layer	YSU
Land Surface	Noah LSM
Initial & Boundary data	ERA5 (0.25° and 6 h)

Ocean	Regional Ocean Modeling System (ROMS) 3.9
Horizontal Resolution | (Domain)	12 km (817×513) / dt = 30 s
Ocean depth	ETOPO2
Vertical Layers	30
Lateral boundary	Mixed Radiation Nudging: temperature, salinity, 3D U and V momentum (baroclinic) Shchepetkin: 2D U and V momentum (barotropic) Chapman_implicit: Free-surface, every 6hours
Boundary Layer Scheme (Vertical Turbulent Mixing)	Generic Length Scale (Umlauf and Burchard, 2003)
Momentum and tracer diffusion	Harmonic
River inflow / Tide	△ / O
Initial & Boundary data	GLORYS12V1 (0.083° / daily)

2-4. Experimental Design

Experiment	Model Configuration	Ocean Initial State	River Discharge	Purpose
UNCPL	**WRF**	-	-	**Atmospheric only**
CPL	**WRF-ROMS**	-	-	**Atmosphere-Ocean** without initialization
CPL_I50	WRF-ROMS	**50-year Ocean spin-up**	-	Atmosphere-Ocean with **Initialization**
CPL_I100	WRF-ROMS	**100-year Ocean spin-up**	-	Atmosphere-Ocean with **Initialization (Long-term)**
CPL_RIV	WRF-ROMS	-	**GloFAS v3.4**	Atmosphere-Ocean with **River Discharge**

3. Results

3-1. Coupled Effect (CPL - UNCPL)

3-2. Spin-up Effect (CPL_I50 / CPL_I100 - CPL)

3-3. River Effect (CPL_RIV - CPL)

3-1. Coupled Effect (CPL - GLORYS)

1993 Monthly | Annual

Monthly & Annual SST/S bias CPL - GLORYS (1993)

Cold SST drift

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Annual
Full	-1.2	-2.1	-2.0	-1.6	-1.1	-0.6	-0.2	+0.1	+0.0	-0.3	-0.8	-1.4	-0.9
BoB	-0.4	-1.3	-1.3	-1.2	-0.9	-0.3	-0.4	-0.2	-0.4	-0.3	-0.7	-1.0	-0.7
N.SCS	-1.0	-2.1	-1.5	-1.2	-1.1	-0.6	-0.6	-0.4	+0.0	-0.4	-0.8	-1.2	-0.9
ECS	-1.1	-1.1	-0.8	-0.7	-0.4	-0.4	-0.6	-0.2	-0.2	-0.6	-0.7	-1.1	-0.7
East Sea	-2.0	-2.7	-2.8	-2.6	-2.2	-1.4	+0.0	+0.4	-0.8	-1.7	-1.6	-2.1	-1.6
Okhotsk	-1.3	-1.7	-1.3	-0.3	+0.7	+1.6	+2.0	+2.4	+2.0	+1.8	+1.4	+1.3	+0.7
Phil. Sea	-1.1	-2.1	-2.3	-2.0	-1.7	-1.2	-0.5	-0.1	+0.2	-0.1	-0.7	-1.4	-1.1
WNP	-1.4	-2.4	-2.4	-2.2	-1.8	-1.2	-0.8	-0.4	-0.3	-0.8	-1.3	-2.0	-1.4

SST bias (°C): 4, 2, 0, −2, −4

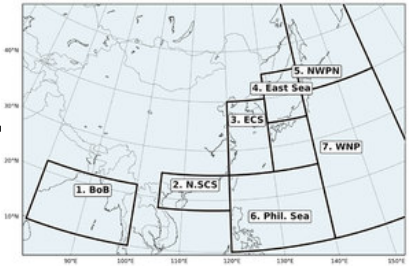
High Salinity bias

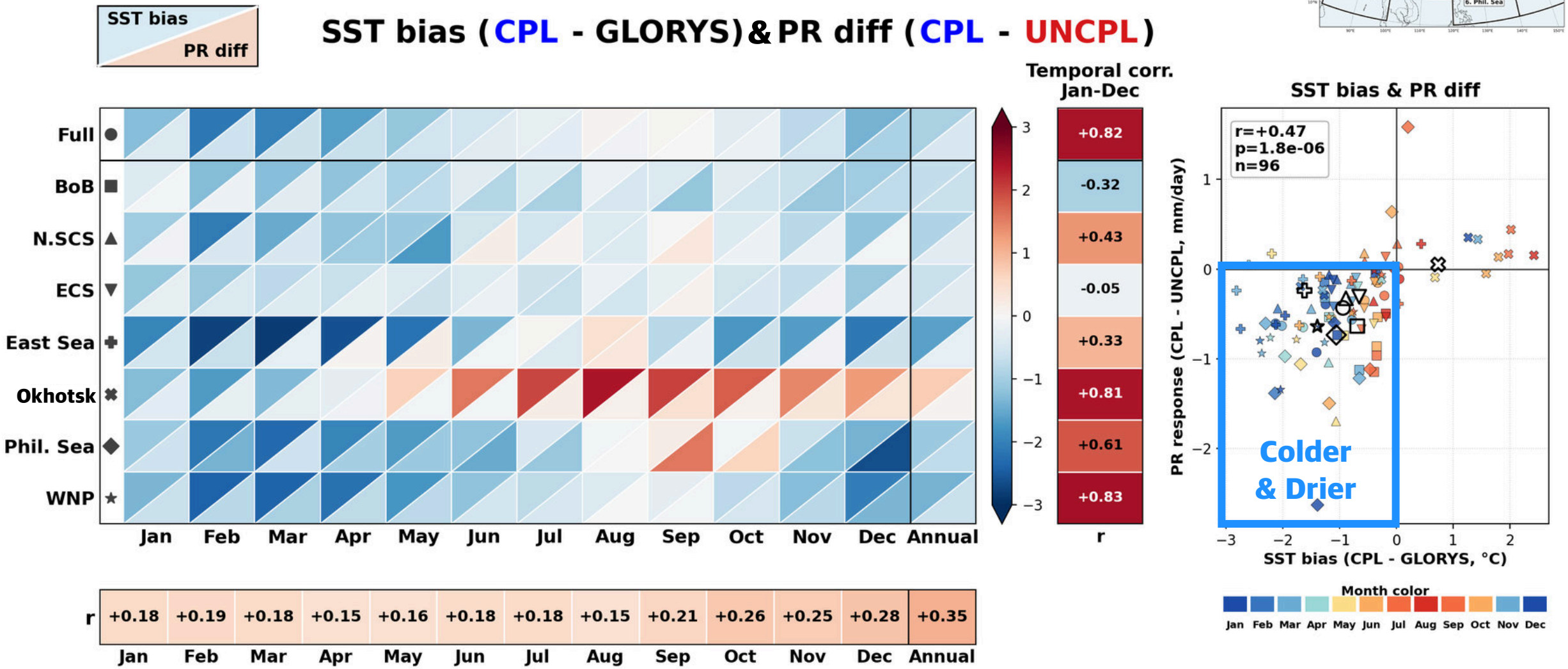
	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Annual
Full	+0.12	+0.20	+0.23	+0.28	+0.35	+0.40	+0.44	+0.48	+0.55	+0.60	+0.59	+0.56	+0.40
BoB	+0.48	+0.71	+0.86	+1.03	+1.21	+1.40	+1.95	+2.50	+3.13	+3.73	+3.76	+3.14	+2.00
N.SCS	+0.09	+0.07	+0.11	+0.20	+0.35	+0.49	+0.74	+0.82	+0.86	+0.92	+1.04	+1.18	+0.57
ECS	+0.19	+0.38	+0.52	+0.73	+0.96	+1.03	+1.15	+1.27	+1.51	+1.54	+1.53	+1.20	+1.00
East Sea	+0.05	+0.06	+0.06	+0.06	+0.07	+0.05	+0.05	+0.16	+0.26	+0.21	+0.16	+0.09	+0.11
Okhotsk	+0.06	+0.12	+0.17	+0.26	+0.44	+0.53	+0.56	+0.56	+0.65	+0.72	+0.71	+0.73	+0.46
Phil. Sea	+0.07	+0.13	+0.17	+0.19	+0.25	+0.32	+0.36	+0.35	+0.33	+0.34	+0.36	+0.28	+0.26
WNP	+0.04	+0.04	+0.02	+0.02	+0.04	+0.07	+0.04	-0.01	-0.04	-0.09	-0.09	-0.04	-0.00

SSS bias (PSU): 1.00, 0.75, 0.50, 0.25, 0.00, −0.25, −0.50, −0.75, −1.00

Values are CPL regional monthly/annual biases relative to GLORYS.

3-1. Coupled Effect (CPL - UNCPL)



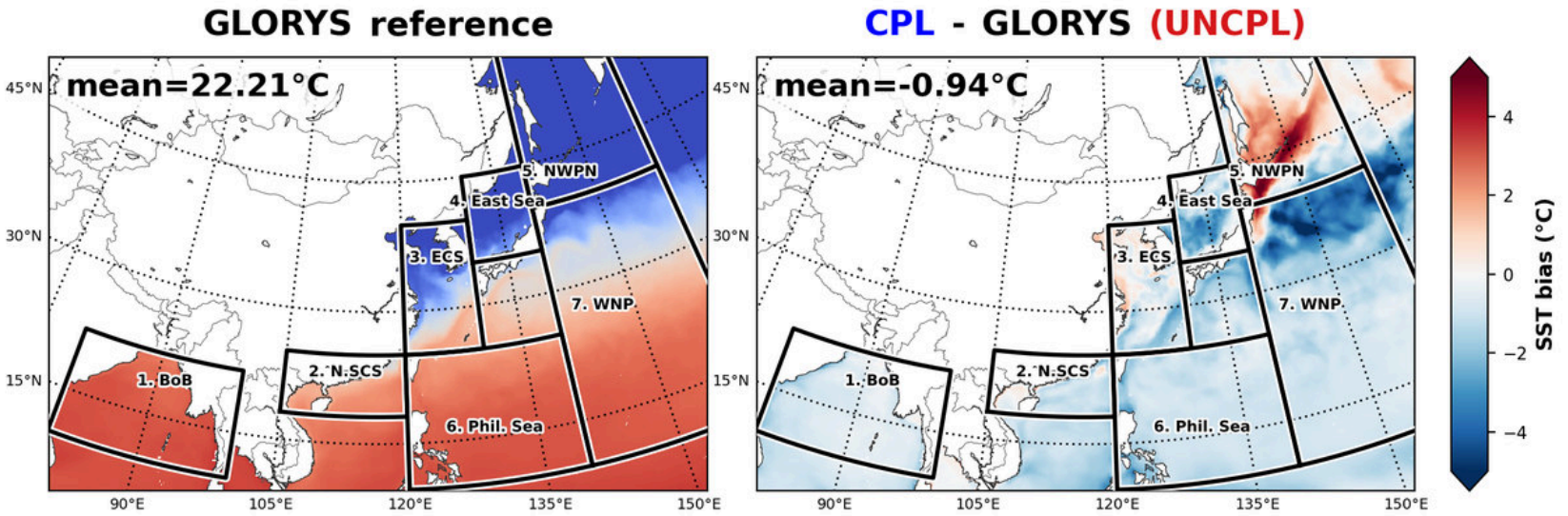


Cell split: upper-left=SST bias, lower-right=PR diff; color midpoint=0, same RdBu_r ±3 scale. Scatter r=+0.47 (p=1.8e-06, n=96).

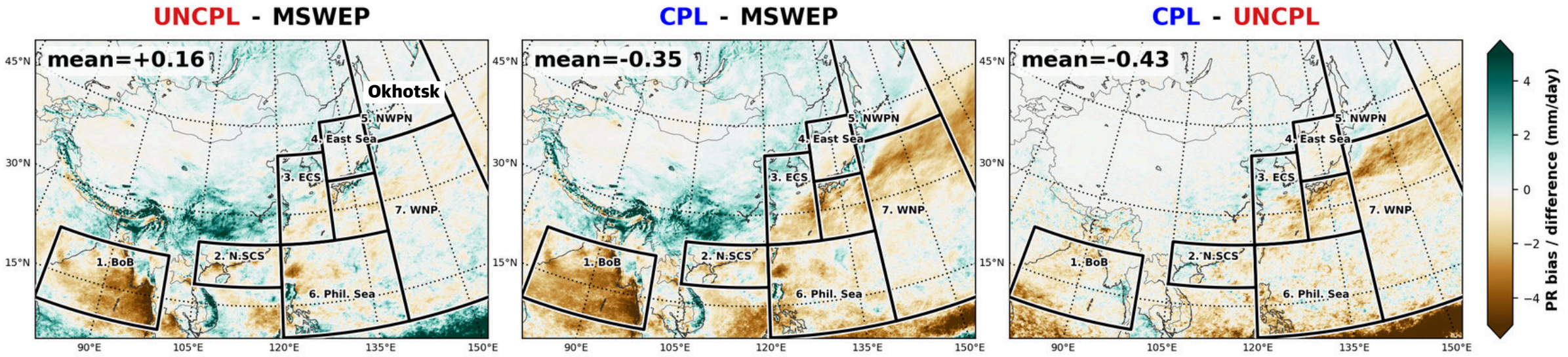
3-1. Coupled Effect (CPL - UNCPL)

1993 Annual

SST



PR

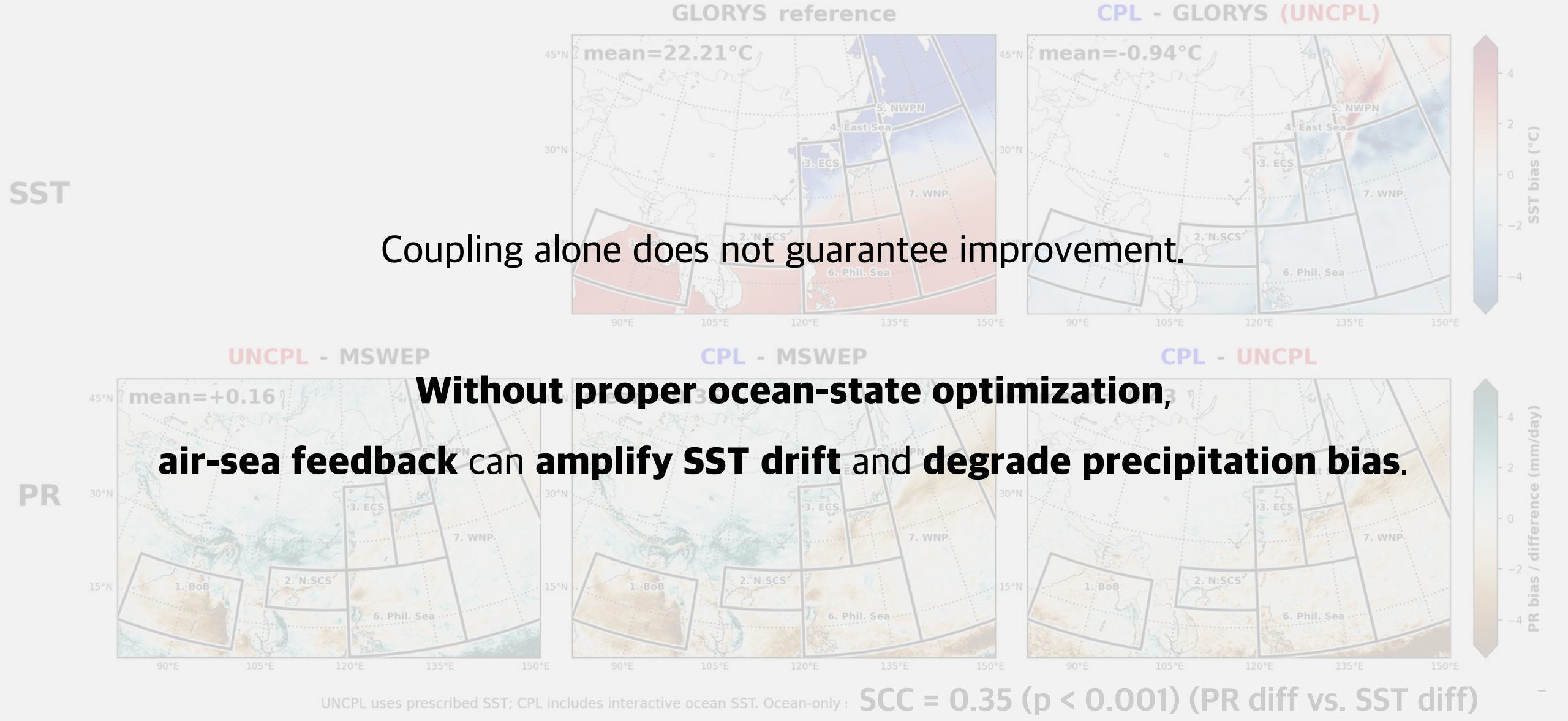


UNCPL uses prescribed SST; CPL includes interactive ocean SST. Ocean-only

SCC = 0.35 (p < 0.001) (PR diff vs. SST diff)

3-1. Coupled Effect (CPL - UNCPL)

1993 Annual



Coupling alone does not guarantee improvement.

Without proper ocean-state optimization,

air-sea feedback can **amplify SST drift** and **degrade precipitation bias**.

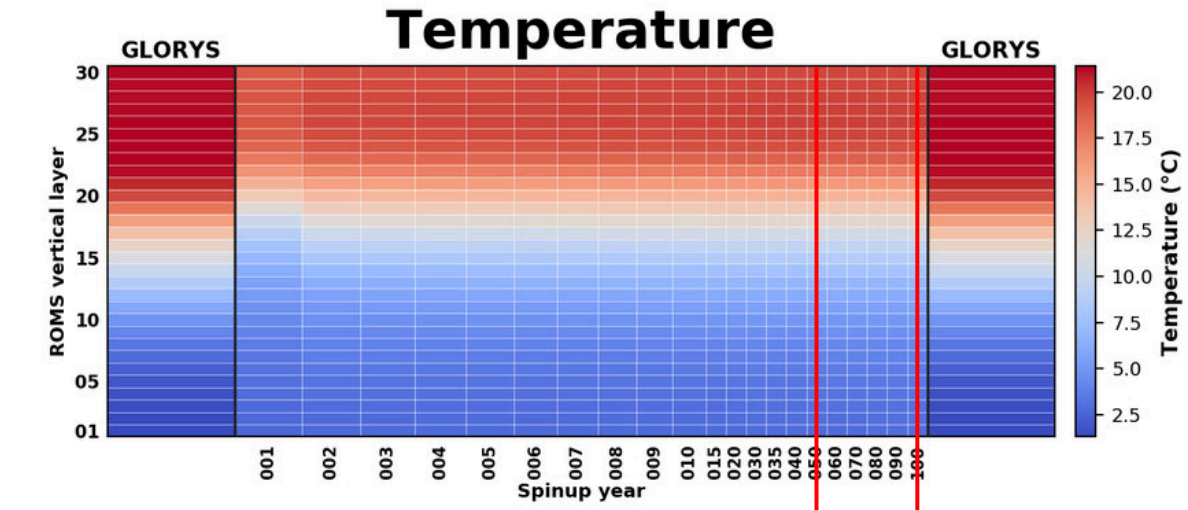
3. Results

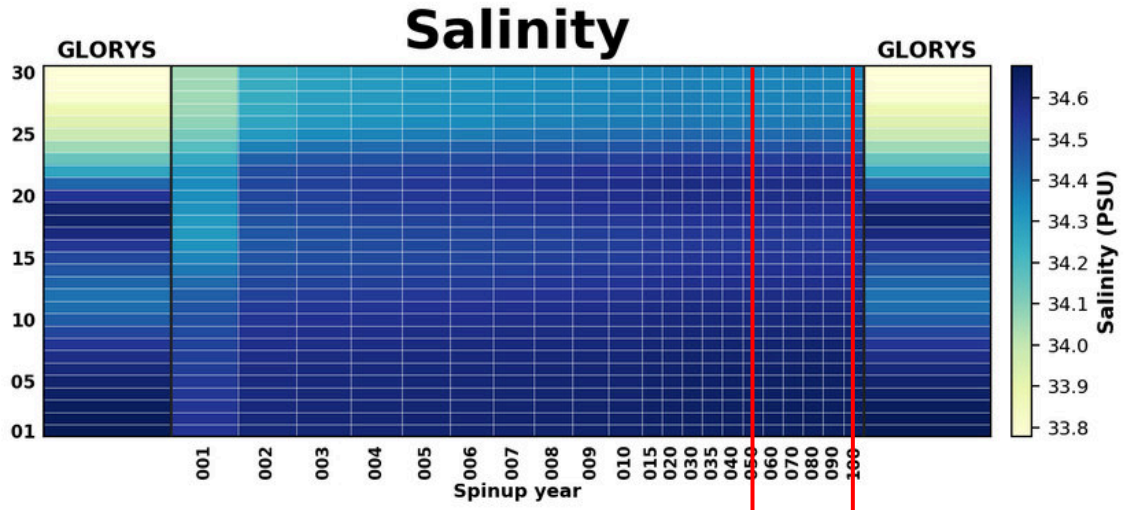
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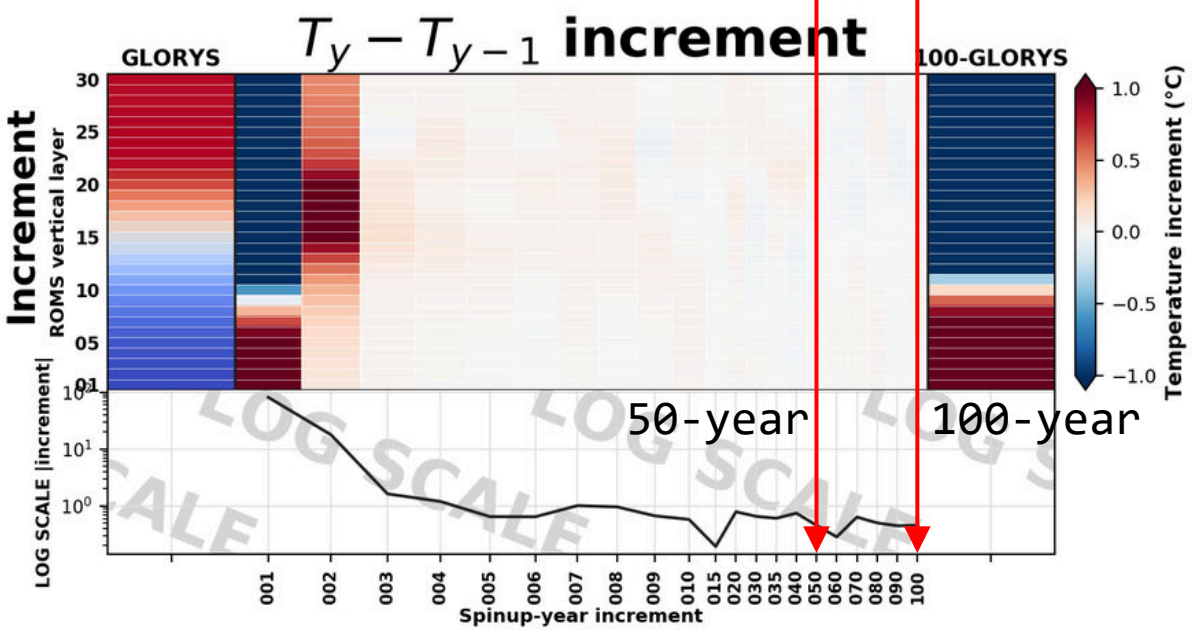
3-2. Spin-up Effect (CPL_I50 / CPL_I100 - CPL)

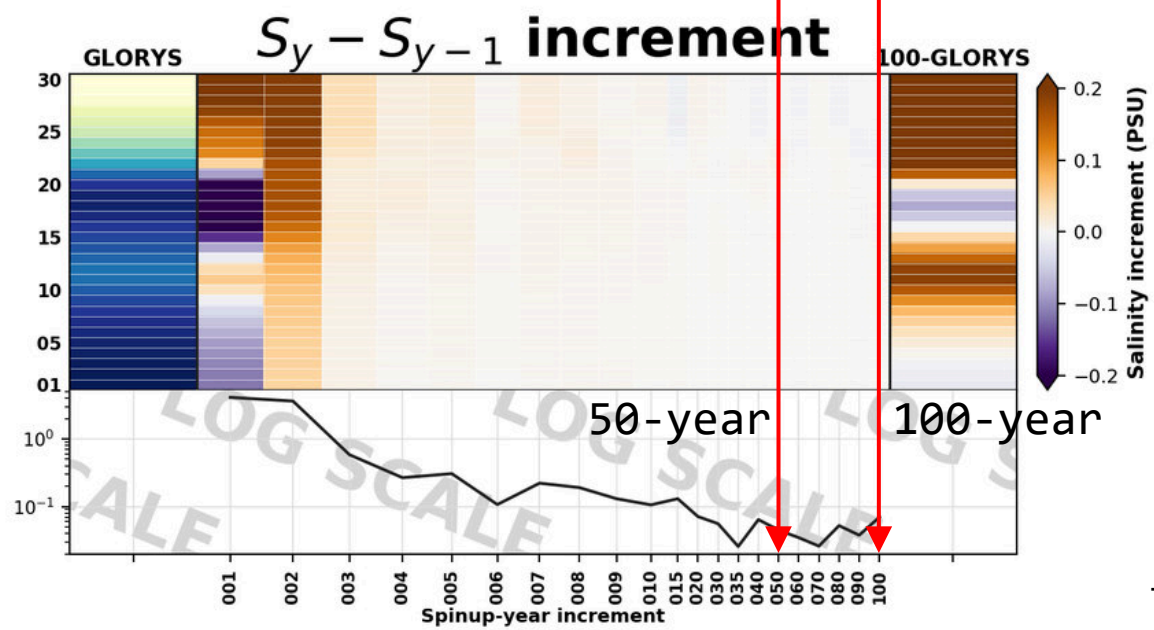
3-3. River Effect (CPL_RIV - CPL)

3-2. Ocean spin-up length sensitivity (CPL_I50 & CPL_I100)





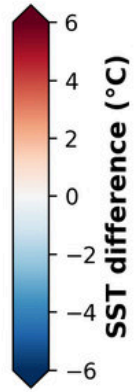




3-2. Ocean spin-up Effect (CPL & CPL_I50) - GLORYS

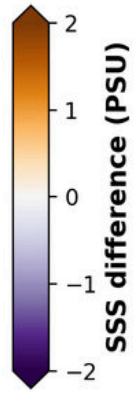
Spin-up Warms Surface Temperature

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Annual
Full	+0.7	+1.5	+1.6	+1.6	+1.4	+1.2	+1.0	+0.8	+0.7	+0.8	+0.9	+1.0	+1.1
BoB	-0.6	-0.3	-0.2	-0.1	-0.1	-0.1	-0.2	-0.3	-0.1	+0.1	+0.2	+0.2	-0.1
N.SCS	-0.1	+1.4	+1.5	+1.4	+0.8	+0.3	+0.3	+0.3	+0.3	+0.6	+0.7	+0.9	+0.7
ECS	+0.3	+0.9	+1.2	+1.3	+1.2	+1.1	+1.0	+0.8	+0.7	+0.8	+0.9	+0.8	+0.9
East Sea	+3.8	+5.2	+5.6	+5.2	+4.3	+4.3	+3.3	+3.0	+2.7	+2.8	+3.4	+4.5	+4.0
Okhotsk	+5.5	+6.3	+6.3	+6.0	+5.3	+4.9	+4.1	+3.2	+2.6	+2.5	+2.6	+3.0	+4.3
Phil. Sea	-0.0	+0.7	+0.8	+0.8	+0.5	+0.3	+0.2	+0.1	+0.1	+0.1	+0.1	+0.2	+0.3
WNP	-0.0	+0.7	+0.8	+0.9	+0.8	+0.7	+0.5	+0.4	+0.3	+0.3	+0.3	+0.3	+0.5



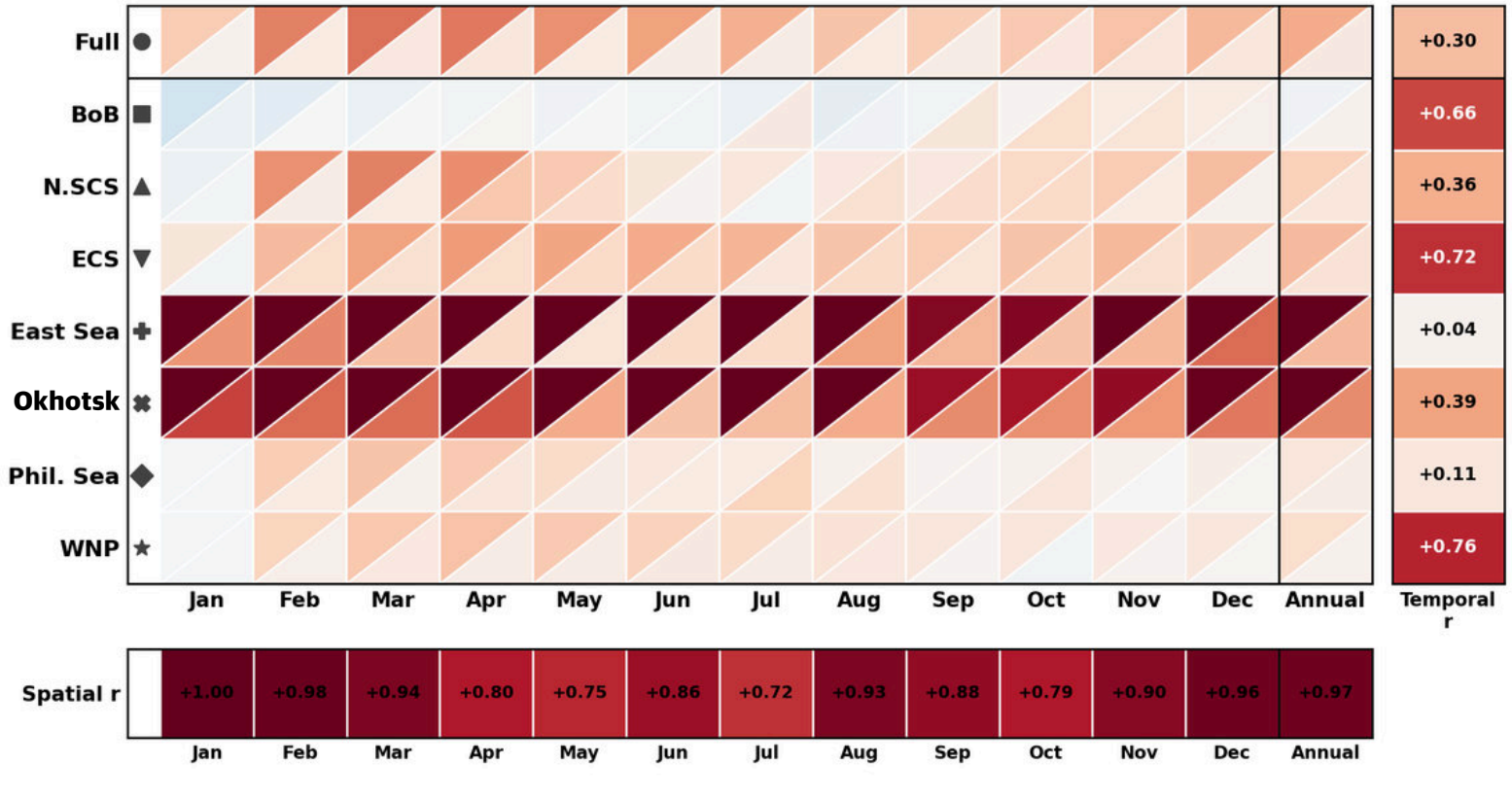
Spin-up Increases Surface Salinity

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Annual
Full	+0.39	+0.34	+0.33	+0.32	+0.33	+0.32	+0.31	+0.30	+0.29	+0.28	+0.26	+0.26	+0.31
BoB	+1.13	+0.70	+0.54	+0.38	+0.27	+0.25	+0.19	+0.17	+0.12	+0.07	+0.04	+0.02	+0.32
N.SCS	+0.61	+0.60	+0.58	+0.56	+0.62	+0.52	+0.37	+0.30	+0.19	+0.11	+0.07	+0.08	+0.38
ECS	+1.78	+1.56	+1.40	+1.32	+1.30	+1.38	+1.40	+1.28	+1.17	+1.06	+1.04	+1.07	+1.31
East Sea	+0.59	+0.55	+0.57	+0.59	+0.64	+0.67	+0.74	+0.77	+0.76	+0.76	+0.71	+0.67	+0.67
Okhotsk	+1.50	+1.37	+1.29	+1.25	+1.22	+1.17	+1.11	+1.02	+0.98	+0.95	+0.88	+0.79	+1.13
Phil. Sea	-0.27	-0.28	-0.25	-0.22	-0.17	-0.15	-0.11	-0.08	-0.05	-0.03	-0.00	+0.01	-0.13
WNP	-0.21	-0.10	-0.02	+0.02	+0.02	+0.02	+0.02	+0.02	+0.04	+0.05	+0.05	+0.06	-0.00

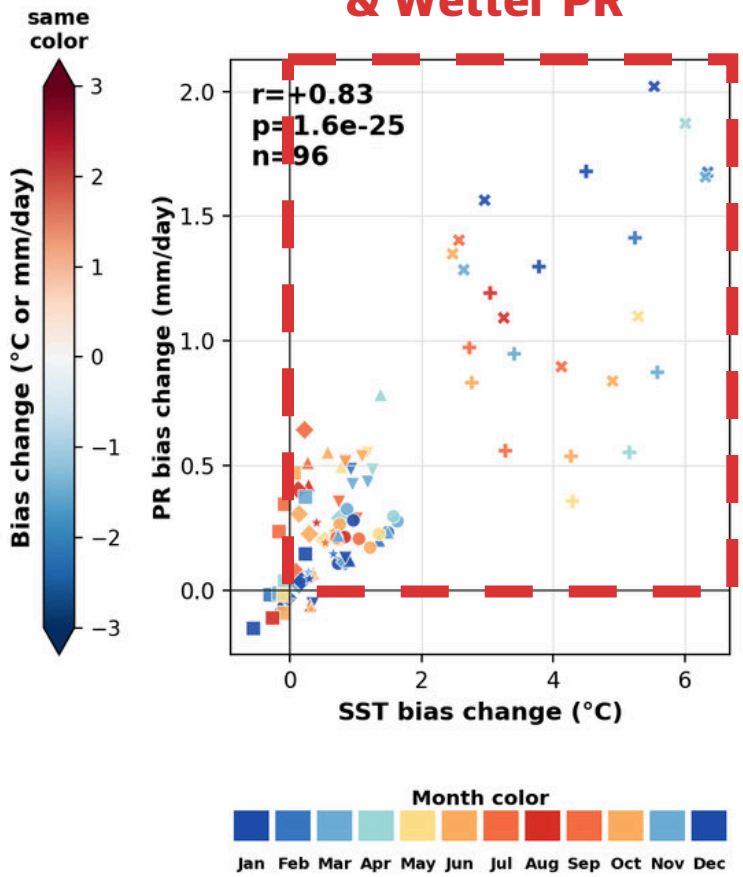


3-2. Ocean spin-up Effect (CPL & CPL_I50) - MSWEP (GLORYS)

Ocean spin-up moves the system
cold & dry state toward → warmer & wetter state



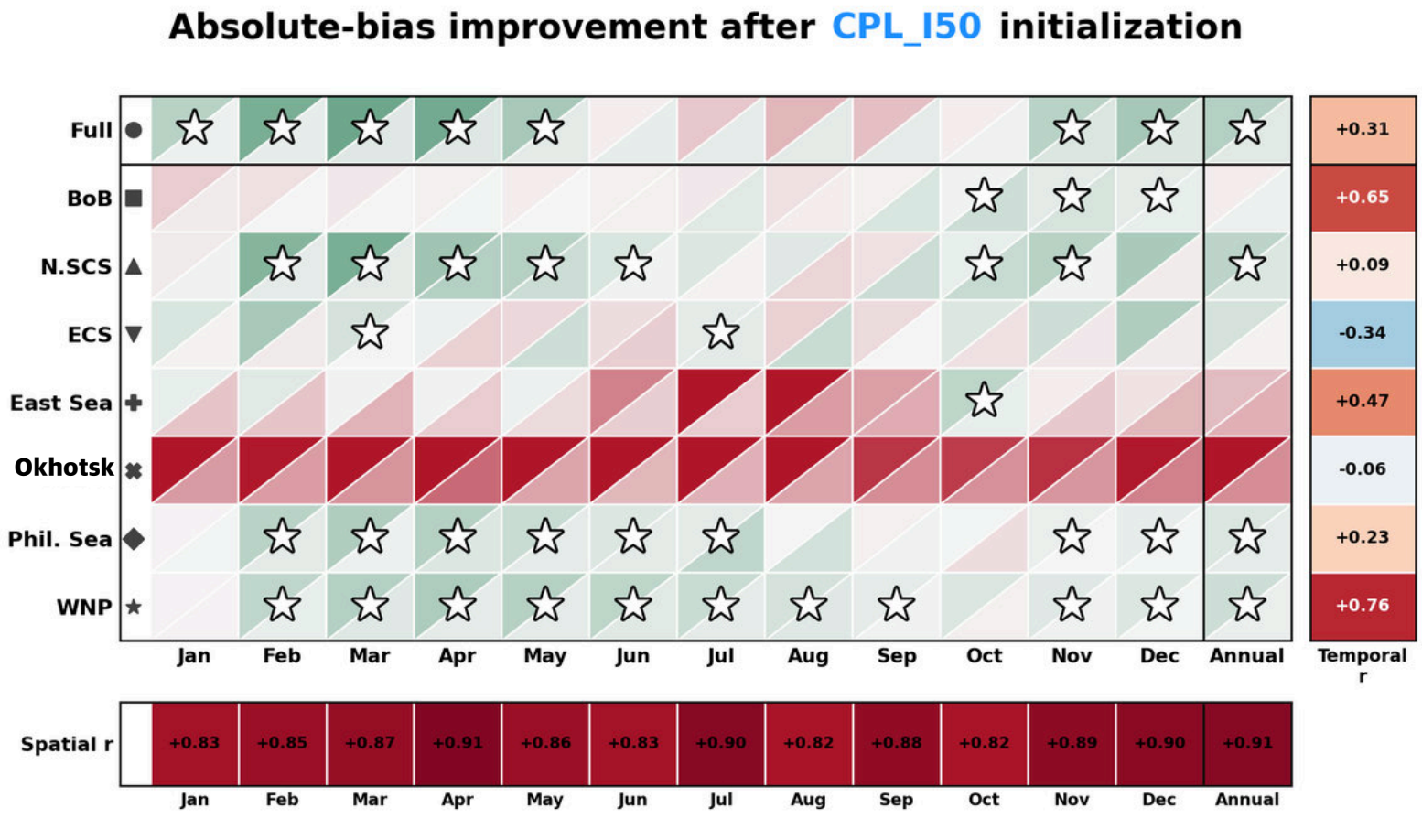
Warmer SST
& Wetter PR

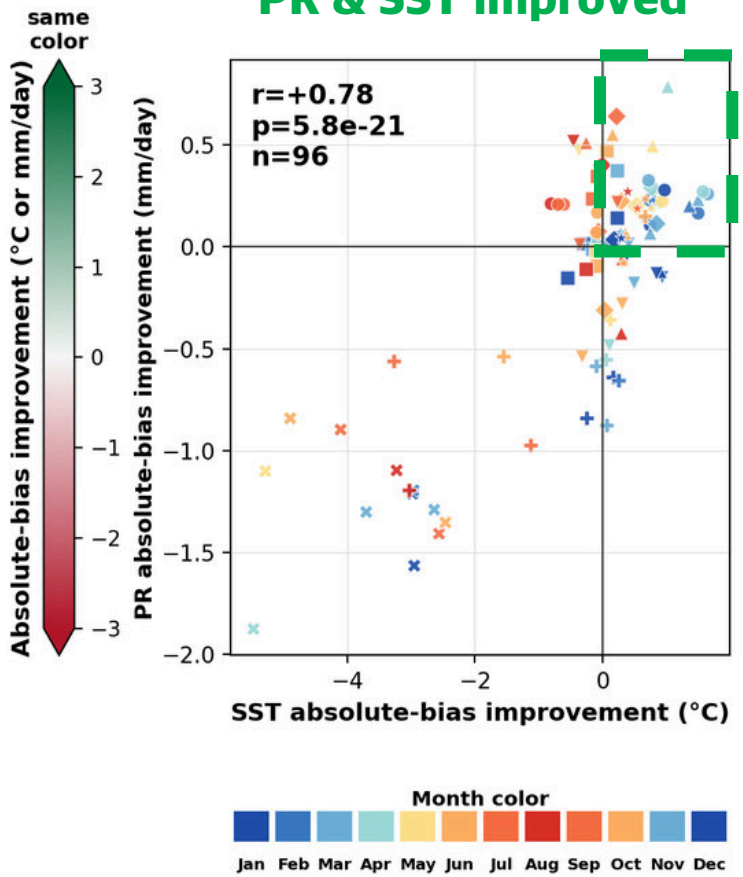


Scatter r=+0.83 (p=1.6e-25, n=96). Spatial r: across regions for each month/annual. Temporal r: Jan-Dec correlation for each region.

3-2. Ocean spin-up Effect (CPL & CPL_I50) - MSWEP (GLORYS)

SST bias and PR response are spatially, temporally correlated

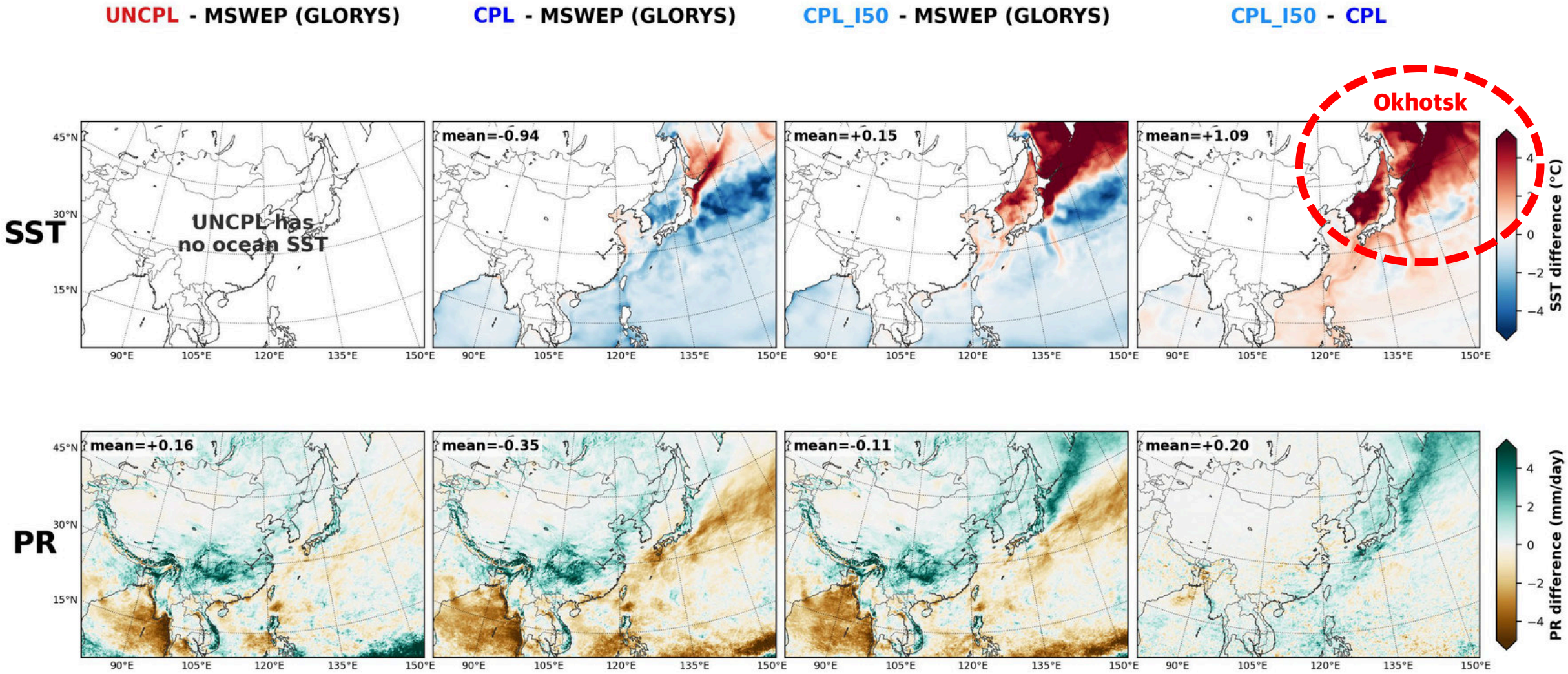




Improved 42/104 (40.4%) **Worsened 62/104 (59.6%)**

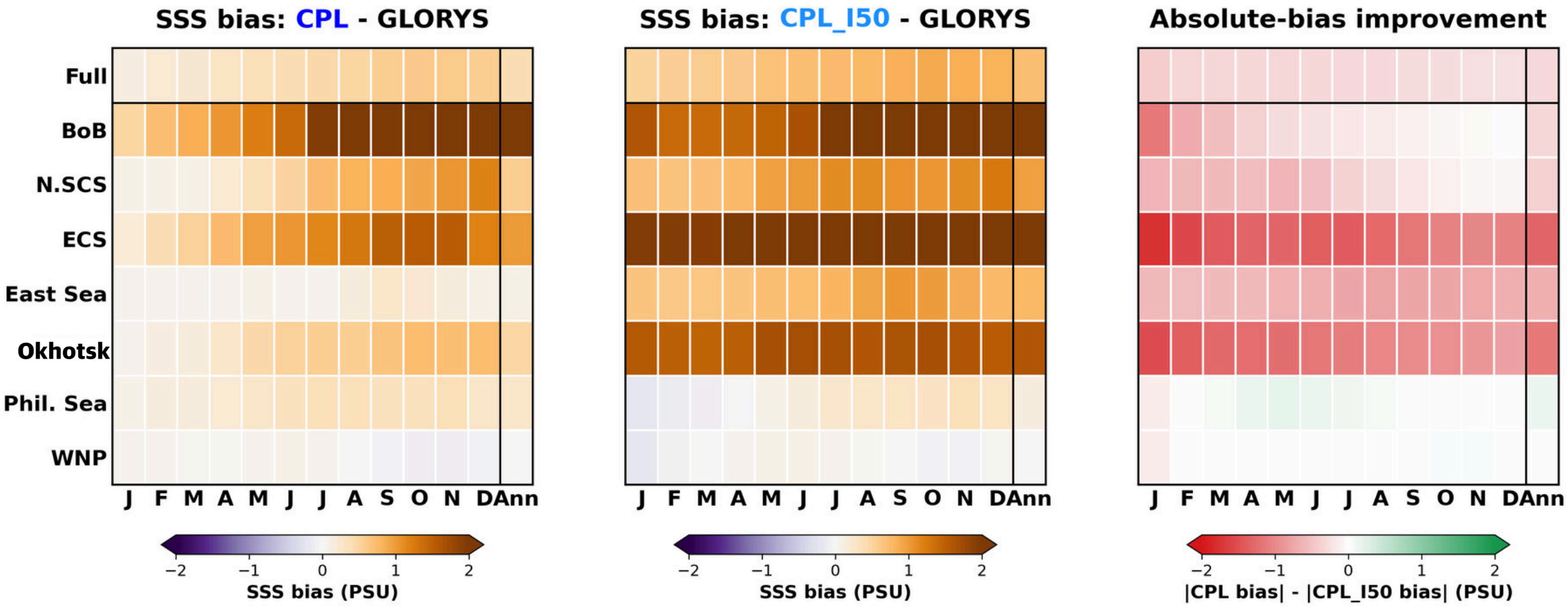
Scatter r=+0.78 (p=5.8e-21, n=96). Spatial r: across regions for each month/annual. Temporal r: Jan-Dec correlation for each region.

3-2. Ocean spin-up Effect (CPL & CPL_I50) - MSWEP (GLORYS)



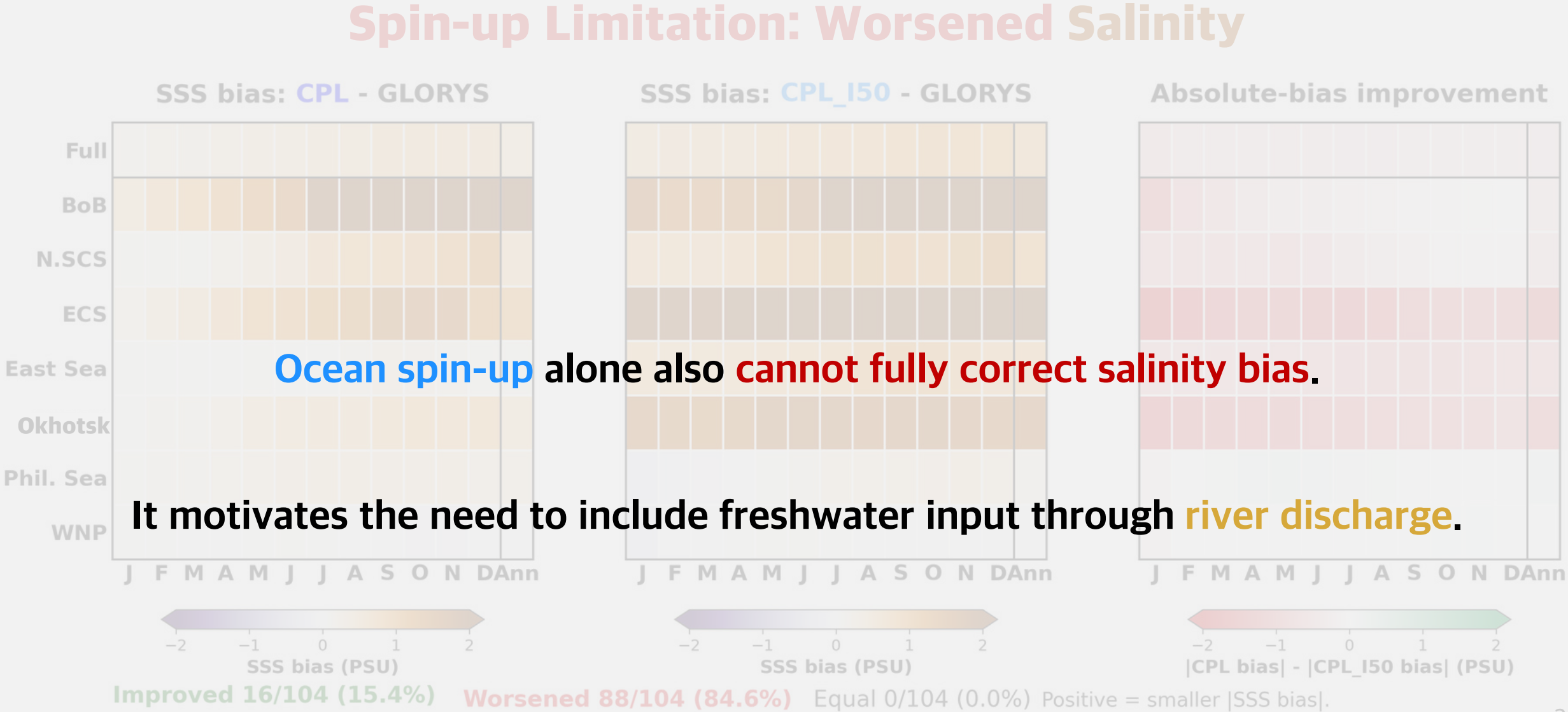
3-2. Ocean spin-up Effect (CPL & CPL_I50) — GLORYS

Spin-up Limitation: Worsened Salinity



Improved 16/104 (15.4%) **Worsened 88/104 (84.6%)** Equal 0/104 (0.0%) Positive = smaller |SSS bias|.

3-2. Ocean spin-up Effect (CPL & CPL_I50) — GLORYS



Ocean spin-up alone also cannot fully correct salinity bias.

It motivates the need to include freshwater input through river discharge.

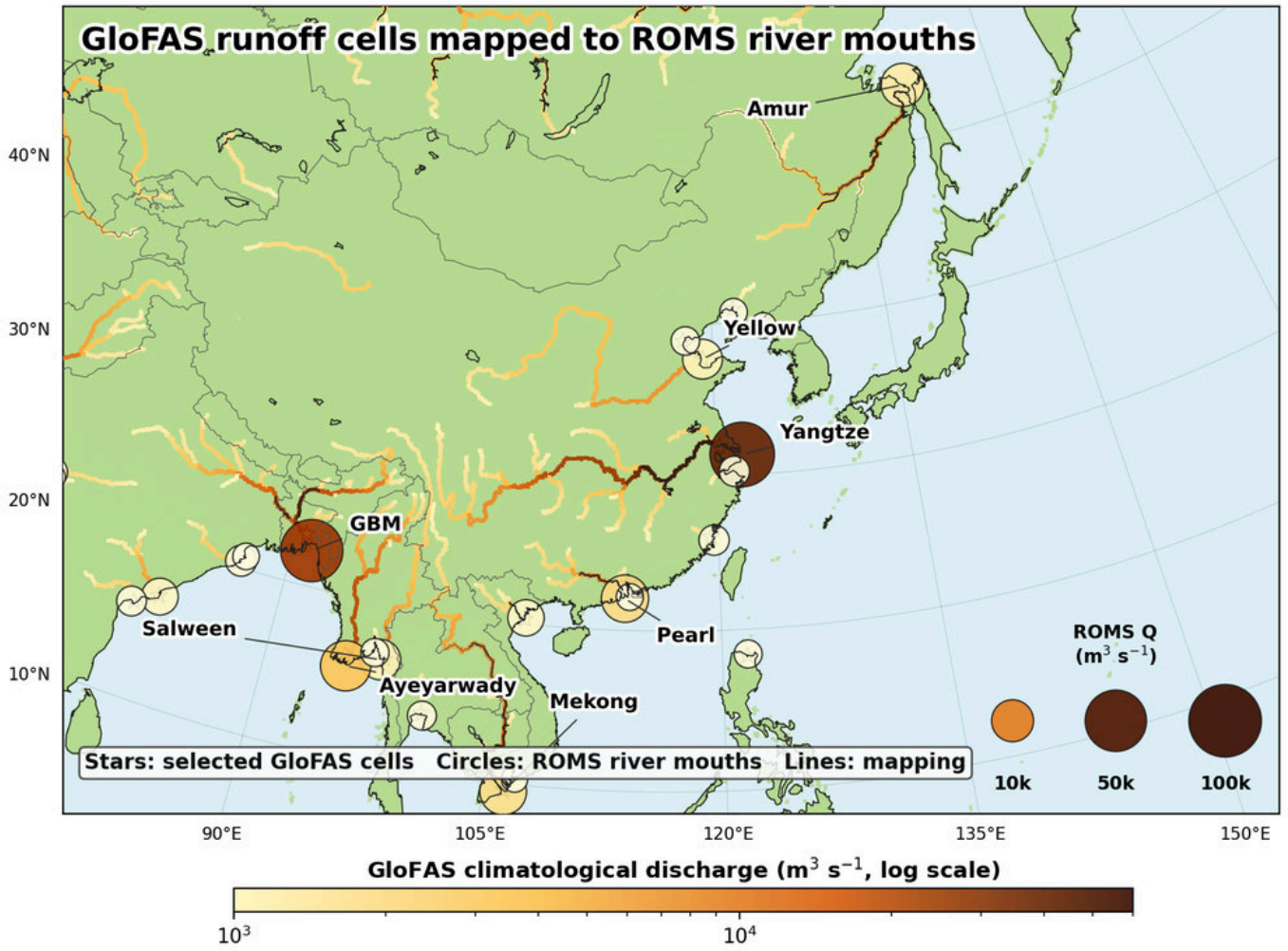
3. Results

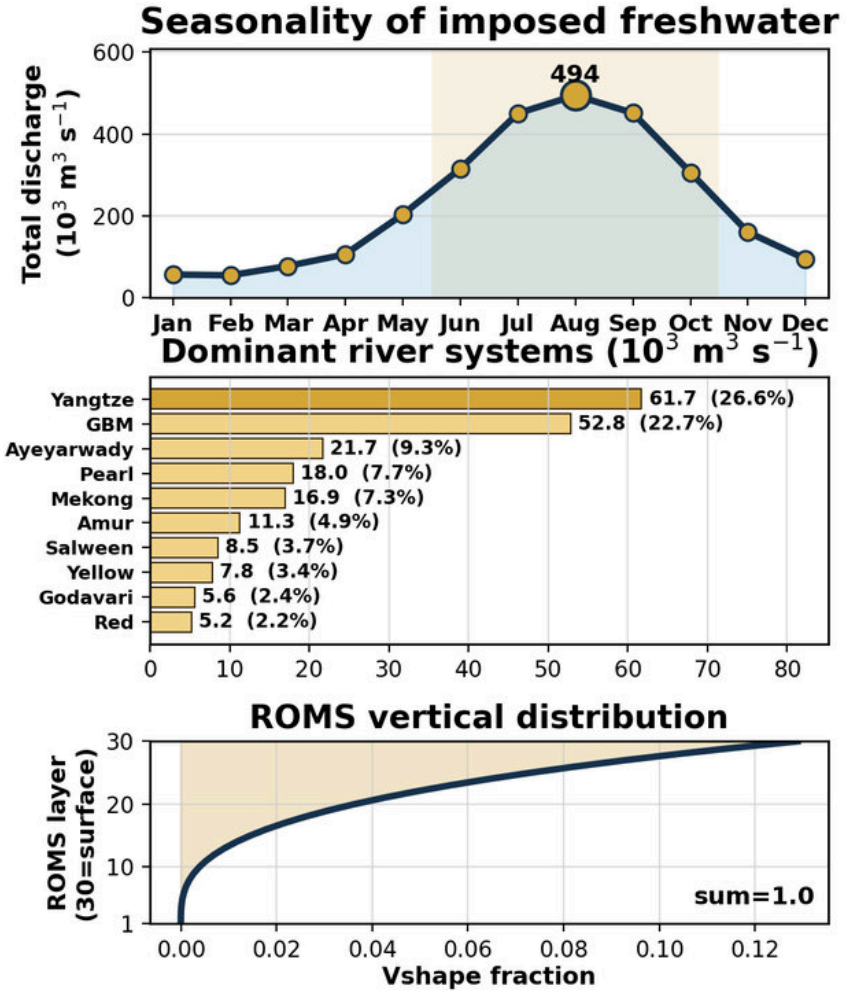
3-1. Coupled Effect (CPL - UNCPL)

3-2. Spin-up Effect (CPL_I50 / CPL_I100 - CPL)

3-3. River Effect (CPL_RIV - CPL)

3-3. From GloFAS Runoff to ROMS River Forcing

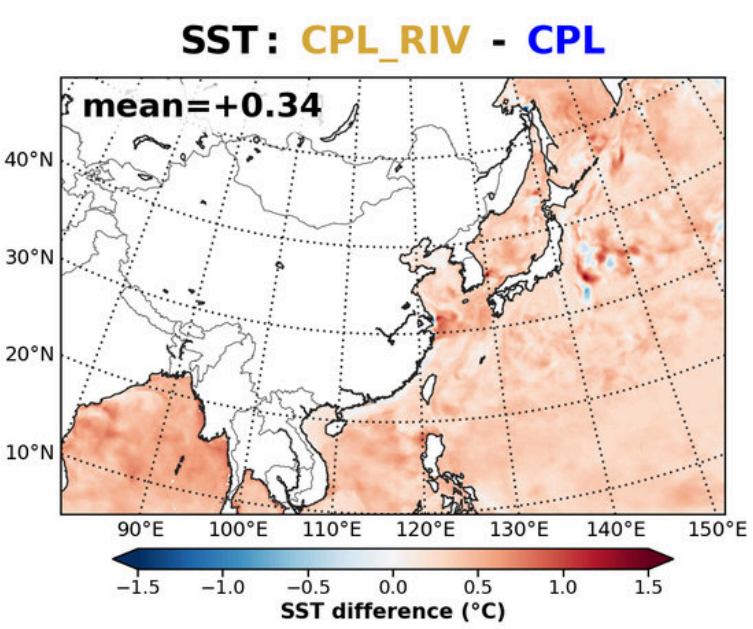


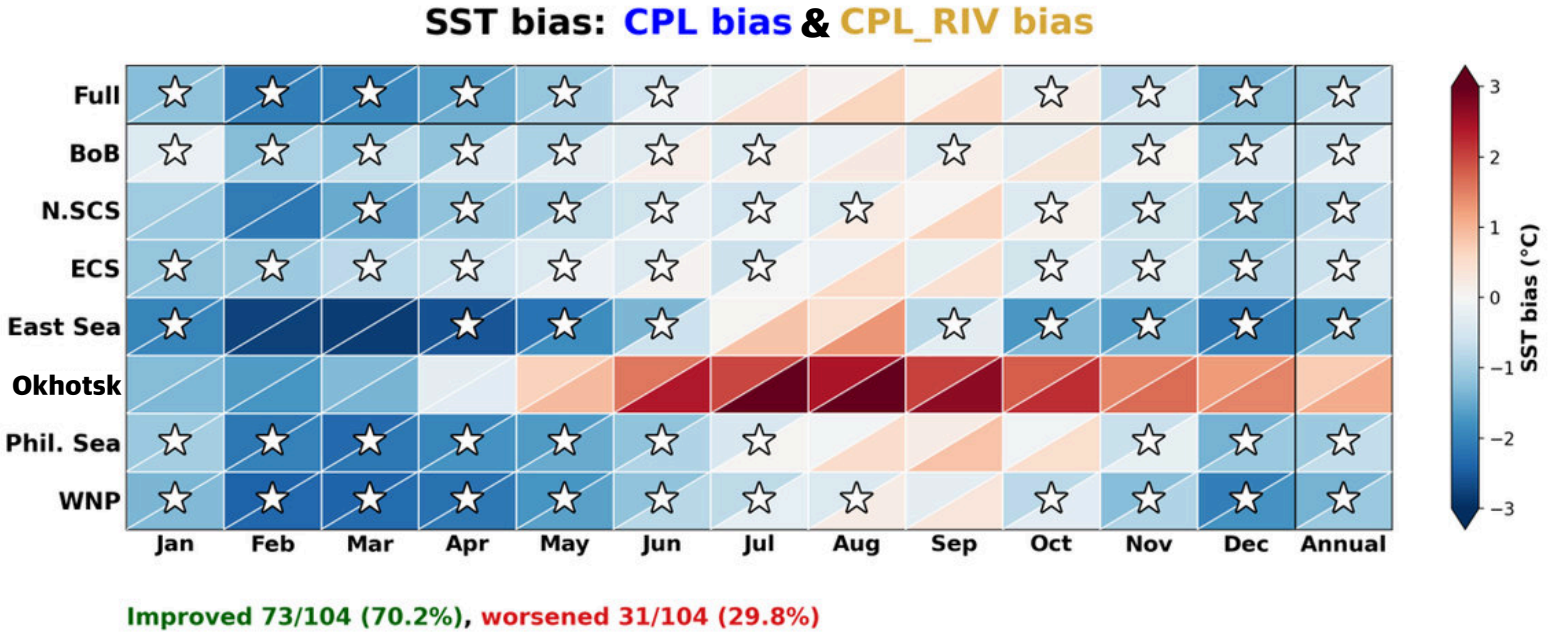


Mouth selection: GloFAS mouth candidates with climatological discharge >=1000 m³ s⁻¹ were mapped to nearest valid ROMS coastal cells.
ROMS forcing: 25 mouths, daily 1993 river_transport, 30-layer Vshape; river_salt=1 PSU; river_temp=-1.6 to 31.5 °C; peak total Q=494×10³ m³ s⁻¹ (Aug).

3-3. Discharge Effect: SST (CPL_RIV — CPL)

Discharge Warms SST



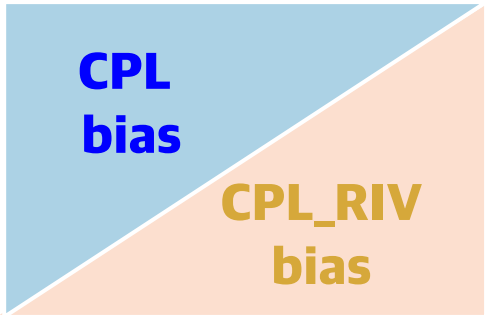


SST annual summary

Domain change: +0.34 °C

|bias| change: -0.34 °C

Improved cells: 73/104 (70.2%)

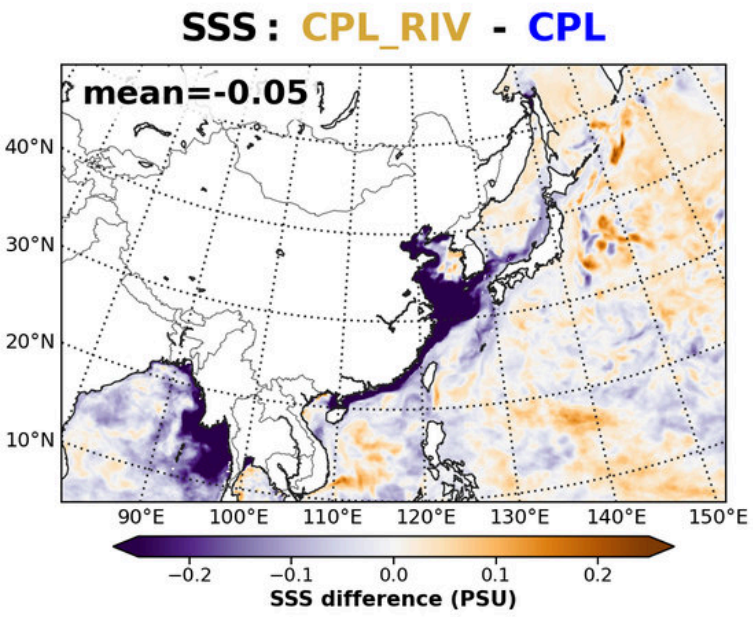


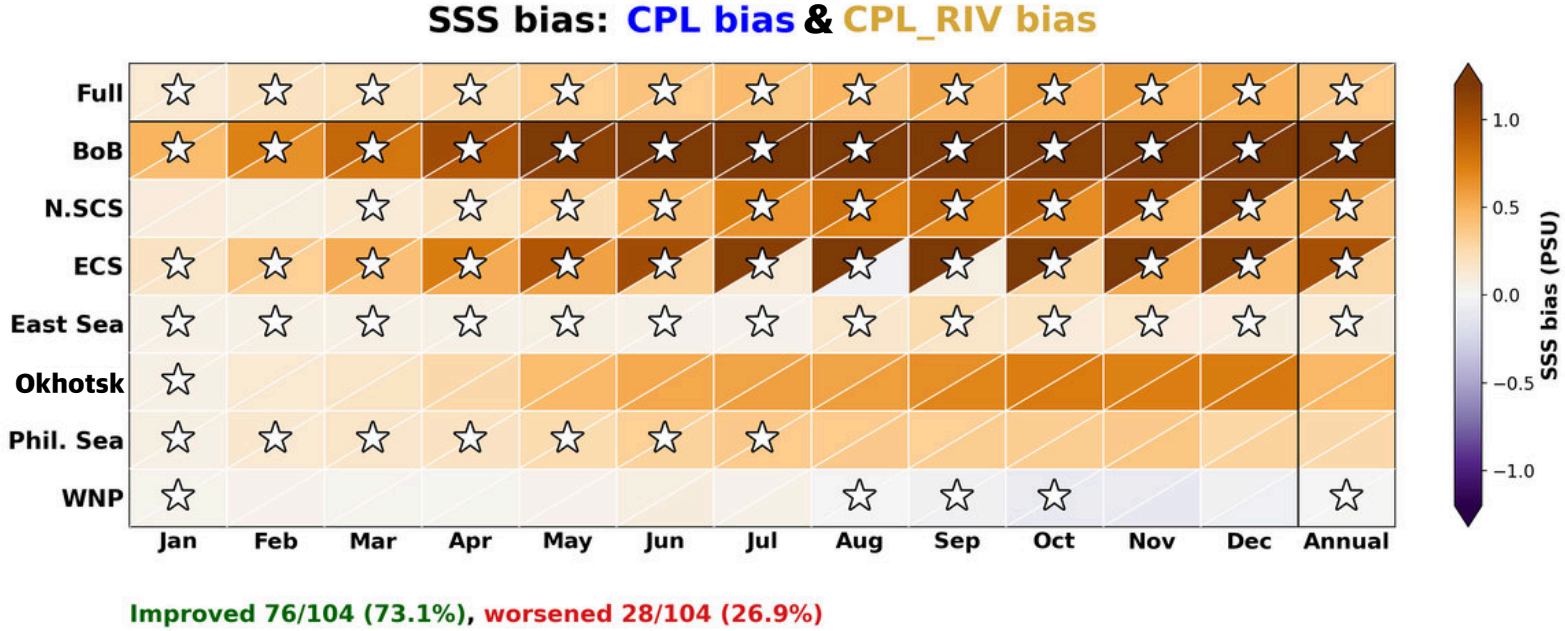
☆ SST bias improved in most cells (70.2%)

Freshwater input modifies the upper-ocean structure & changes the surface thermal response

3-3. Discharge Effect: SSS (CPL_RIV — CPL)

Direct effect to Salinity





Improved 76/104 (73.1%), worsened 28/104 (26.9%)

SSS annual summary

Domain change: -0.05 PSU

|bias| change: -0.05 PSU

Improved cells: 76/104 (73.1%)

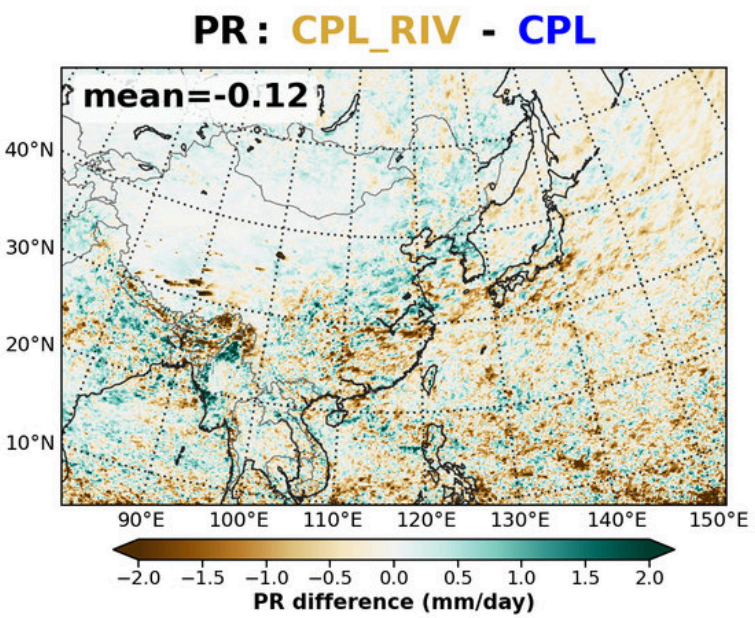


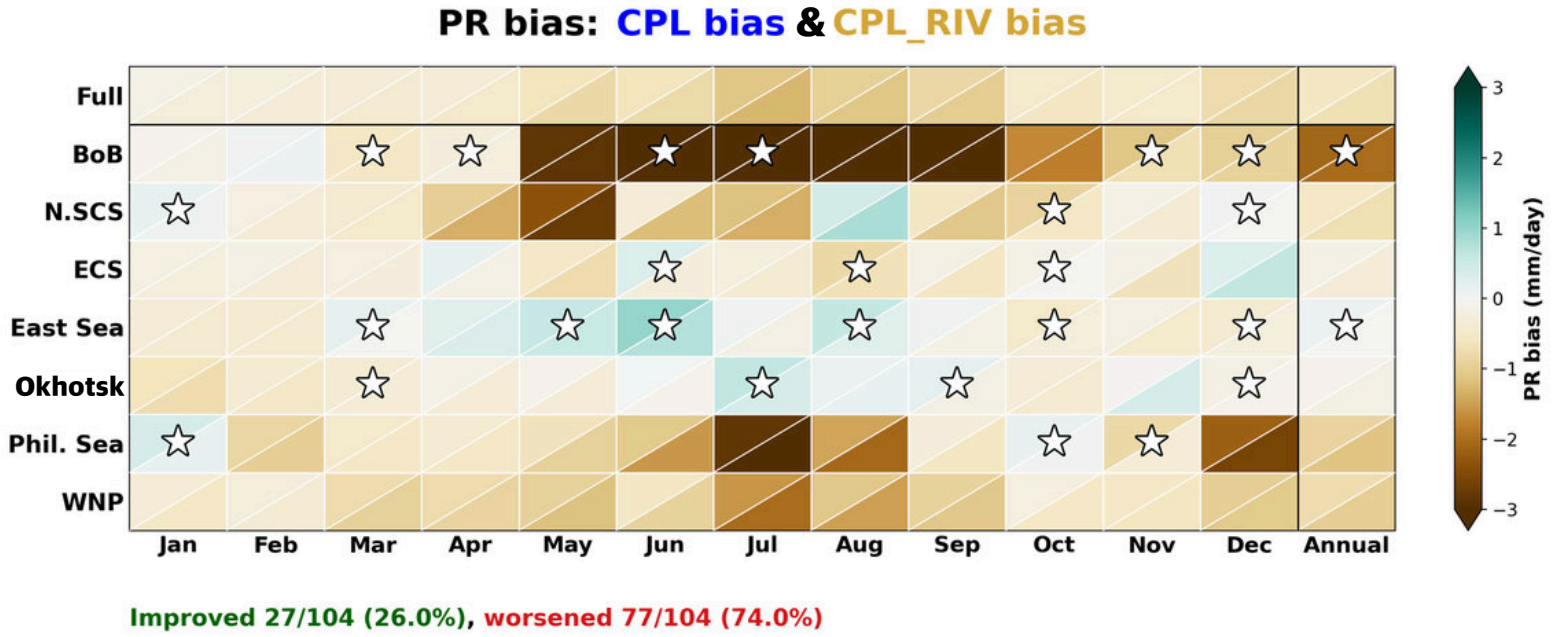
☆ SSS bias improved in most cells (73.1%)

Freshwater lowers SSS relative to CPL.
More improved in coastal regions
& freshwater-influenced areas.

3-3. Discharge Effect: PR (CPL_RIV — CPL)

PR Improved only in a limited cells





Improved 27/104 (26.0%), worsened 77/104 (74.0%)

PR annual summary

Domain change: -0.12 mm/day

|bias| change: +0.12 mm/day

Improved cells: 27/104 (26.0%)

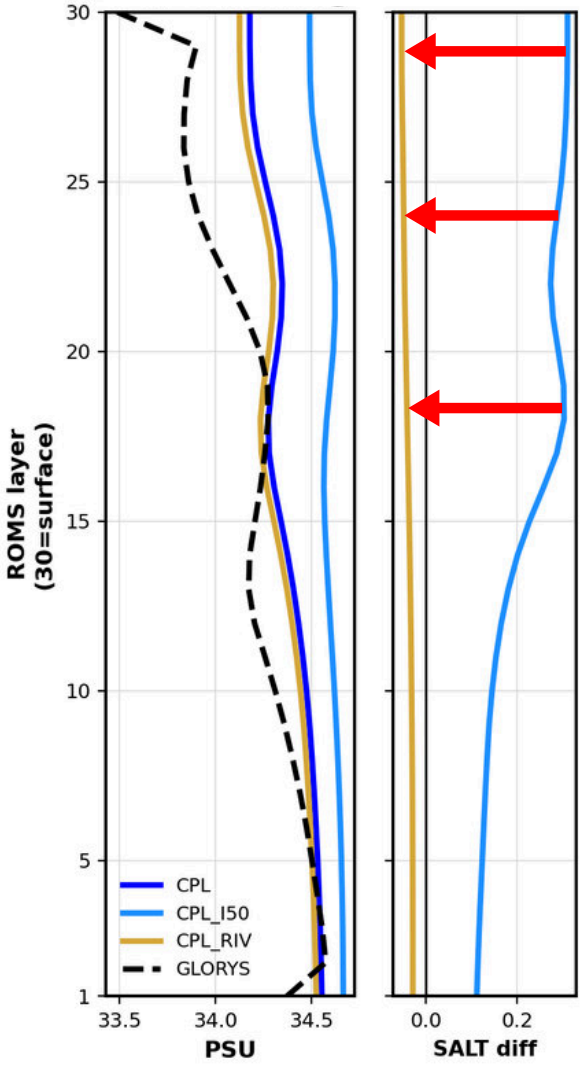


☆ PR bias worsened in most cells (74.0%)

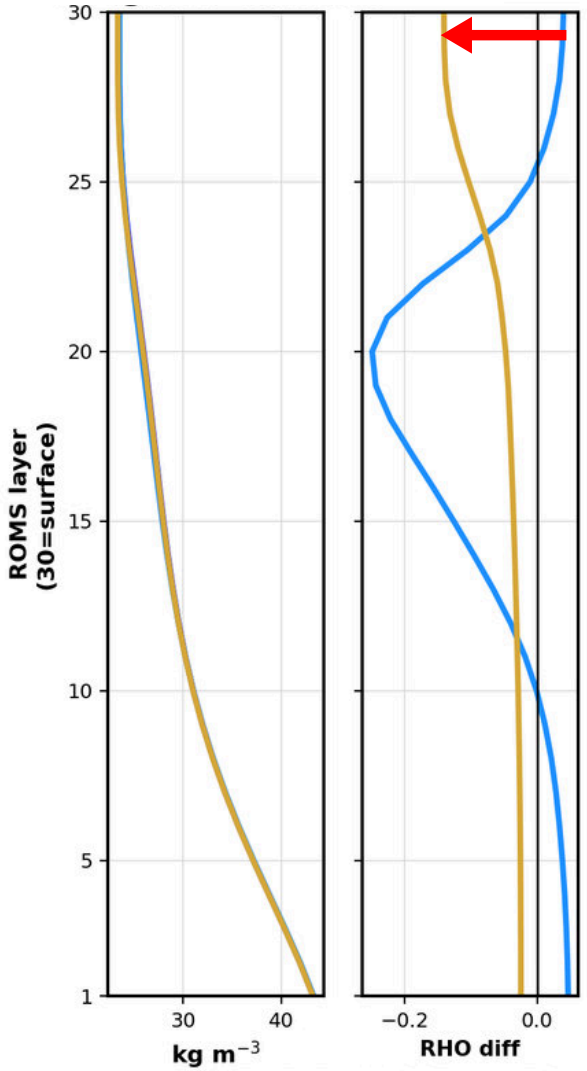
Improved ocean surface state does not directly guarantee better precipitation.

3-3. Discharge Effect - Stratification

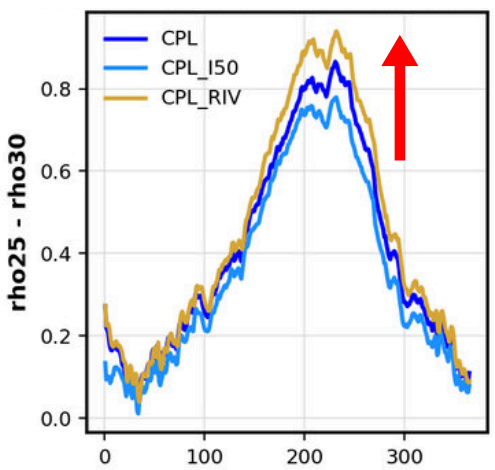
1 Lower surface salinity



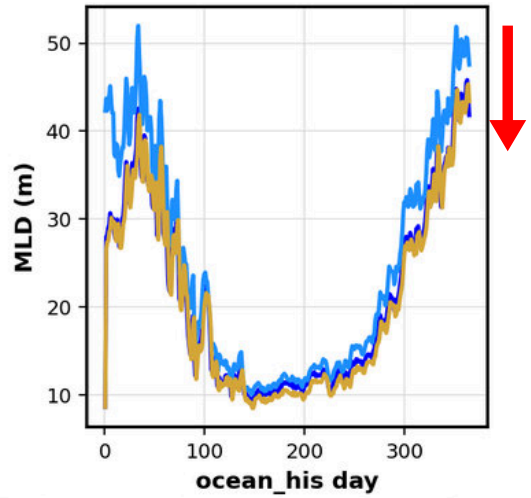
2 Lighter Surface



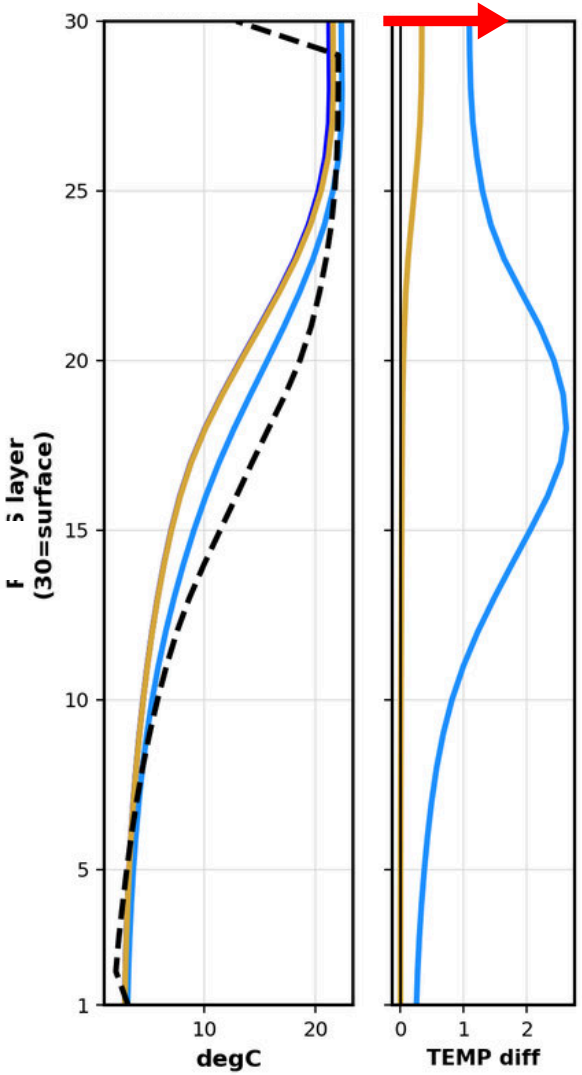
3 Stronger Stratification



Shallower MLD



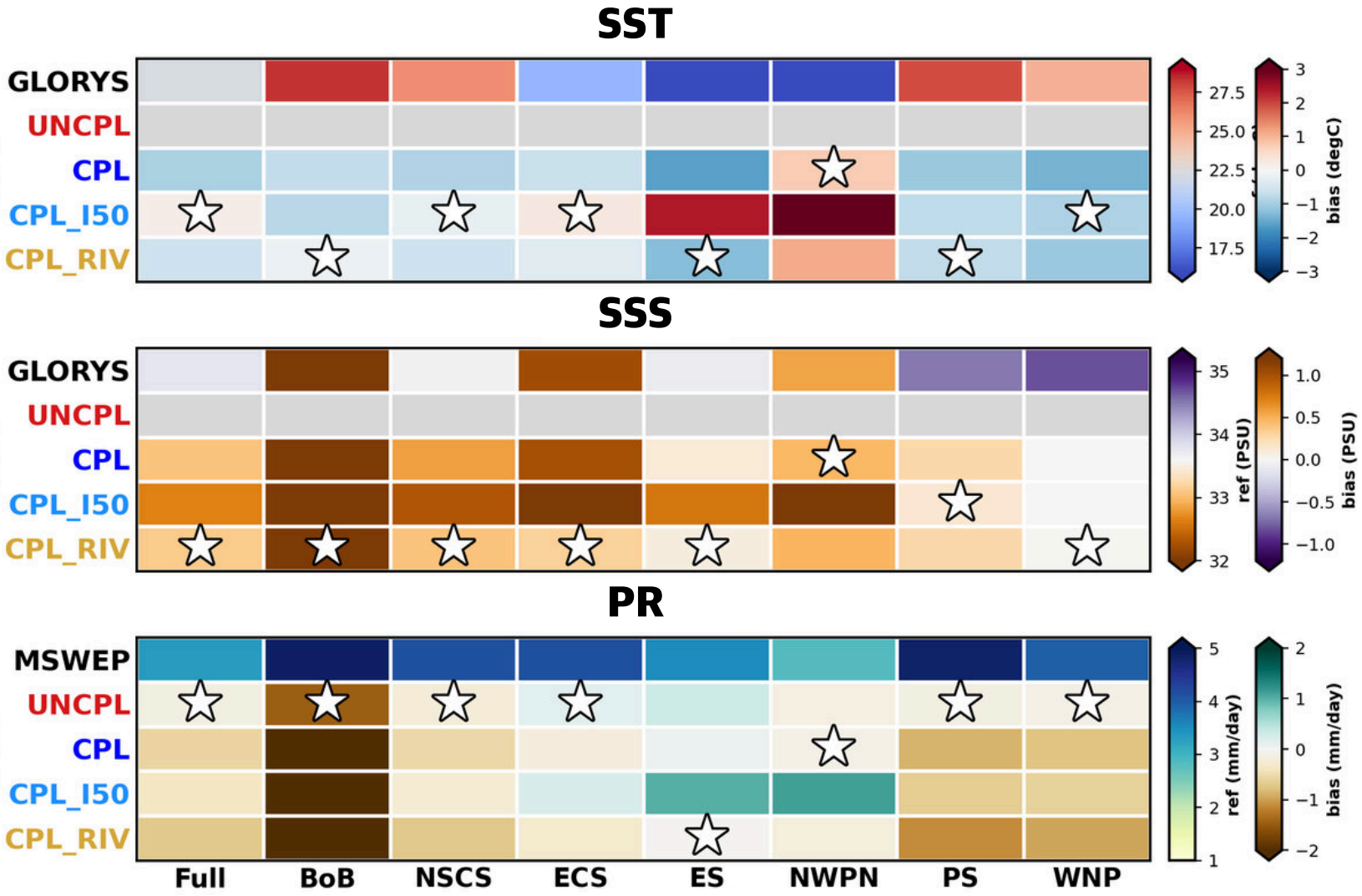
4 Warmer Surface



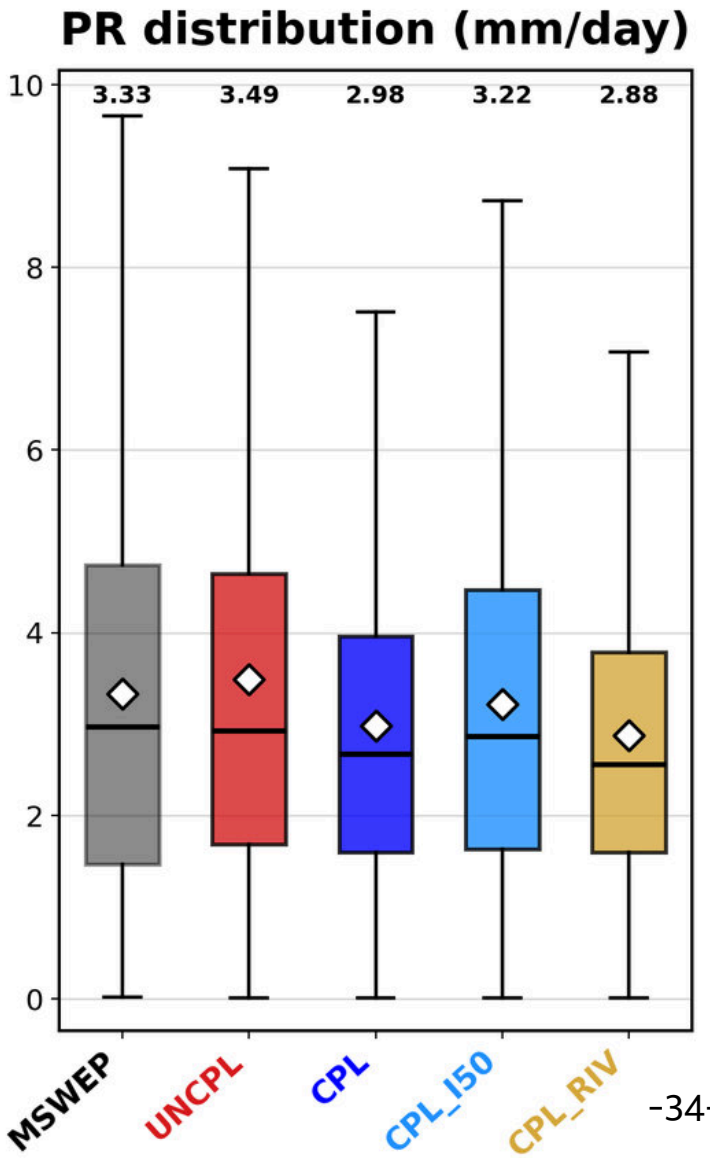
Full-domain RIV-CPL: SSS=-0.05 PSU, rho25-rho30=+0.037, MLD=-0.6 m, SST=+0.34 degC.

4. Summary

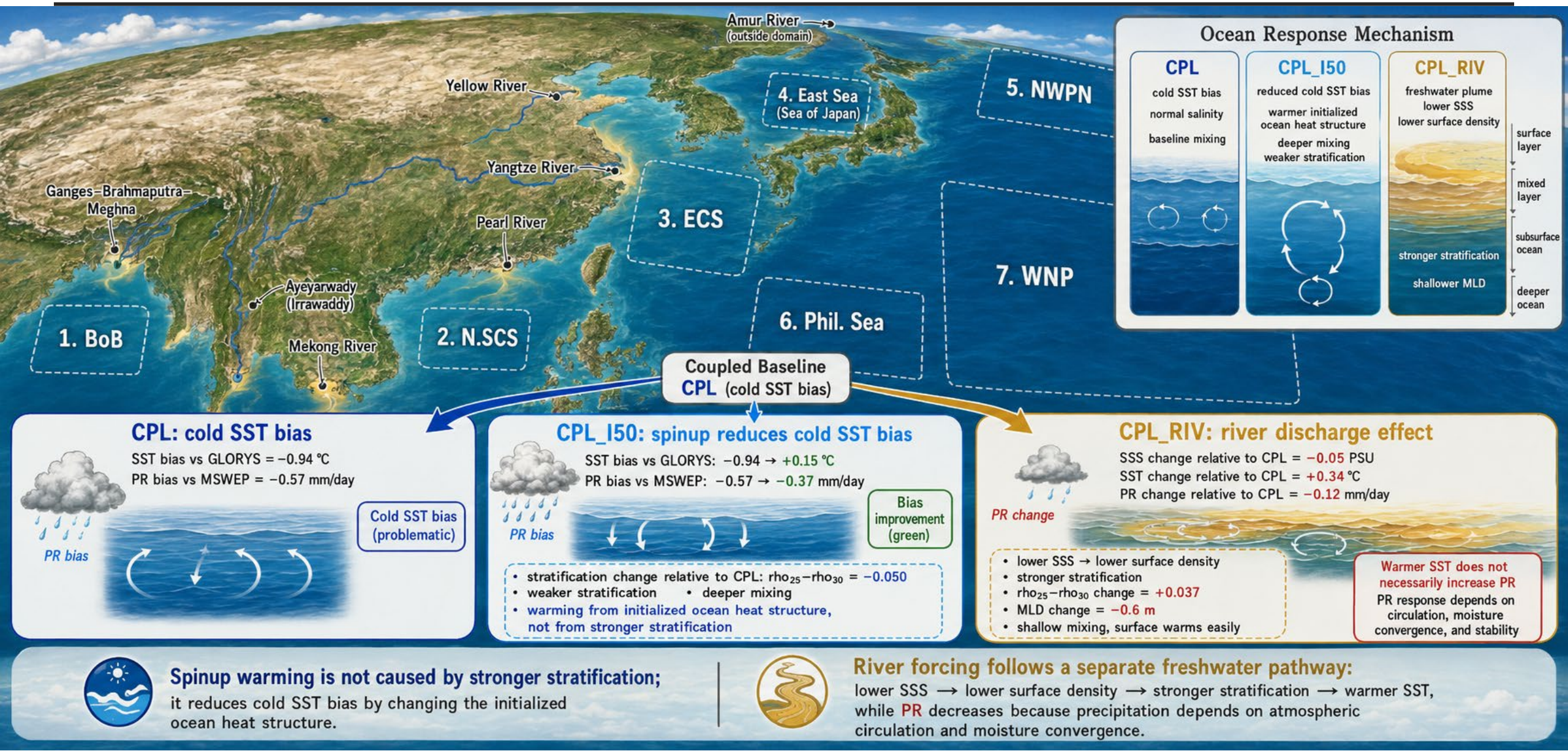
4. Summary



Reference row: absolute value. Model rows: bias relative to reference. Star: smallest |bias|.



4. Summary


Amur River
(outside domain)
Yellow River
4. East Sea
(Sea of Japan)
5. NWPN
Yangtze River
Ganges–Brahmaputra–
Meghna
3. ECS
Pearl River
7. WNP
Ayeyarwady
(Irrawaddy)
1. BoB
2. N.SCS
6. Phil. Sea
Mekong River
Ocean Response Mechanism
CPL
cold SST bias
normal salinity
baseline mixing
CPL_I50
reduced cold SST bias
warmer initialized
ocean heat structure
deeper mixing
weaker stratification
CPL_RIV
freshwater plume
lower SSS
lower surface density
stronger stratification
shallower MLD
surface layer
mixed layer
subsurface ocean
deeper ocean
Coupled Baseline
CPL (cold SST bias)
CPL: cold SST bias
SST bias vs GLORYS = −0.94 ℃
PR bias vs MSWEP = −0.57 mm/day
PR bias
Cold SST bias
(problematic)
CPL_I50: spinup reduces cold SST bias
SST bias vs GLORYS: −0.94 → +0.15 ℃
PR bias vs MSWEP: −0.57 → −0.37 mm/day
PR bias
Bias
improvement
(green)
• stratification change relative to CPL: rho25−rho30 = −0.050
• weaker stratification • deeper mixing
• warming from initialized ocean heat structure,
not from stronger stratification
CPL_RIV: river discharge effect
SSS change relative to CPL = −0.05 PSU
SST change relative to CPL = +0.34 ℃
PR change relative to CPL = −0.12 mm/day
PR change
• lower SSS → lower surface density
• stronger stratification
• rho25−rho30 change = +0.037
• MLD change = −0.6 m
• shallow mixing, surface warms easily
Warmer SST does not
necessarily increase PR
PR response depends on
circulation, moisture
convergence, and stability
Spinup warming is not caused by stronger stratification;
it reduces cold SST bias by changing the initialized
ocean heat structure.
River forcing follows a separate freshwater pathway:
lower SSS → lower surface density → stronger stratification → warmer SST,
while PR decreases because precipitation depends on atmospheric
circulation and moisture convergence.

4. Future Work

1. **Spin-up + River Forcing Experiment**
   Apply river discharge to the spin-up ocean state (CPL_I50_RIV) to test whether salinity improvement and reduced cold SST drift can coexist.
   Also, the spin-up with river discharge included.

2. **Sea-Ice Process Improvement**
   Include or improve sea-ice treatment to reduce the persistent NWPN cold bias, especially near the Okhotsk Sea and subarctic western North Pacific.

3. **Atmospheric Response Diagnosis**
   Diagnose IVT, moisture convergence, vertical motion, and lower-tropospheric stability to explain why warmer SST does not necessarily increase precipitation.

4. **Ocean Structure Validation**
   Compare MLD, stratification, and vertical T/S profiles with GLORYS by region and season to evaluate whether the initialized ocean structure is physically realistic.

5. References

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Thank you

Contact: junseo1026@unist.ac.kr

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